

The Impact of Binary Stripping of Massive Stars on Core-Collapse Supernovae – I. Explodability and Nucleosynthesis

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Accepted XXX. Received YYY; in original form ZZZ

ABSTRACT

Binary interaction can substantially alter massive-star evolution and, in turn, pre-supernova structure, potentially changing core-collapse outcomes and nucleosynthetic yields. We present a systematic comparison of explodability and nucleosynthesis in single-star and binary-stripped progenitors using the parameterised, spherically symmetric PUSH explosion framework. We employ sets of non-rotating, solar-metallicity single-star and binary-stripped progenitors from [Farmer et al. \(2023\)](#), spanning zero-age main-sequence (ZAMS) masses from 11–45 $M_{\odot}$, and perform explosion simulations and detailed nucleosynthesis post-processing for successful explosions. We find that binary stripping alters the explodability landscape but does not produce a systematic shift towards either successful or failed explosions. Successfully exploding single-star and binary-stripped models occupy broadly similar regions of explosion parameter space, with explosion energies of ~ 0.65 – 1.3 B and neutron star remnant masses of ~ 1.5 – 2.0 $M_{\odot}$. Binary stripping produces the largest differences in nucleosynthetic yields at low ZAMS masses, particularly for material undergoing explosive O and Si burning. Using a Salpeter initial mass function to weight the explodability landscape of both populations, we find that both occupy broadly similar elemental yield patterns, with most elements differing by only $\sim 20\%$ with K showing the largest difference, with an enhancement factor of ~ 3.4 in the binary-stripped population. The predicted ^{56}Ni yields and iron-group abundances are broadly consistent with observations of core-collapse supernovae and the metal-poor star HD 84937. Together these results highlight that binary stripping can alter individual explosion outcomes and nucleosynthetic yields, but that the predicted explodability landscape remains strongly dependent on the adopted explosion treatment.

Key words: Radiative transfer – Transients: supernovae – Methods: numerical - Stars: binaries

1 INTRODUCTION

Core-collapse supernovae (CCSNe) mark the energetic deaths of massive stars. At lower zero-age main-sequence mass ($M_{\text{ZAMS}} \simeq 8$ – $10 M_{\odot}$), they may arise from the collapse of an ONeMg core, while more massive stars undergo collapse after forming an Fe core. During these explosions, newly synthesised material is ejected into the surrounding environment, enriching the interstellar medium with intermediate-mass and iron-group elements. CCSNe therefore provide an important source of chemical enrichment and feedback, and play a central role in the chemical evolution of galaxies.

Most massive stars do not evolve in complete isolation. A large fraction are found in systems with multiple stars ([Abt 1983](#); [Mason et al. 2009](#)), and many are expected to undergo at least one phase of mass transfer during their lifetimes ([Sana et al. 2012](#); [Moe & Di Stefano 2017](#)). This binary interaction can fundamentally alter the evolutionary pathway of a stellar system ([Podsiadlowski et al. 1992](#); [Brown et al. 1996](#); [Wellstein & Langer 1999](#); [Eldridge et al. 2008](#);

[Langer 2012](#); [de Mink et al. 2013](#)). During mass transfer, much of the outer hydrogen-rich envelope may be removed from the primary star, with some fraction of this material accreted by the companion. This changes the subsequent stellar evolution, the final pre-collapse structure, and the chemical yields of the system ([Braun & Langer 1995b](#); [Laplace et al. 2021](#); [Farmer et al. 2021, 2023](#)).

Despite the importance of binarity, explosive nucleosynthesis in binary-stripped progenitors has been less extensively explored ([Braun & Langer 1995a](#); [Vartanyan et al. 2021](#); [Farmer et al. 2023](#); [Boccioli & Roberti 2026](#)) than for single-star progenitors (e.g., [Woosley et al. 1993](#); [Heger et al. 2001, 2002](#); [Heger & Woosley 2002](#); [Woosley & Heger 2007](#); [Pignatari et al. 2016](#); [Limongi & Chieffi 2018](#); [Curtis et al. 2019](#); [Ebinger et al. 2020a](#); [Andrews et al. 2020](#); [Sieverding et al. 2023](#)). This is partly because the CCSN explosion mechanism itself remains a long-standing theoretical challenge. A central limitation has been the availability of suitably detailed binary progenitor models evolved through the final stages of nuclear burning to the point of core collapse. Evolving binary progenitors through these final stages requires sufficiently detailed nuclear reaction networks, as electron capture reactions can substantially alter the

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structure and composition, while even relatively small differences in the resulting pre-collapse structure can affect the subsequent core-collapse outcome (Farmer et al. 2016; Vartanyan et al. 2018; Laplace et al. 2021). The computational cost of these larger reaction networks has therefore made it difficult to produce extensive grids of binary progenitors evolved self-consistently to core collapse (Laplace et al. 2021).

The difficulty of modelling CCSNe does not end with producing suitably detailed pre-collapse progenitor grids, as the explosion mechanism itself also remains challenging to model. It requires the coupled treatment of hydrodynamics, gravity, nuclear equations of state, neutrino transport, nuclear reaction rates, and possibly neutrino flavour conversion. Although major progress has been made in multidimensional simulations over the last decade, fully self-consistent calculations remain computationally expensive and are not yet practical for population-level studies. Although a growing number of 2D and 3D simulations now follow the evolution from collapse through bounce and shock revival (e.g. Janka 2012; Burrows 2013; Nakamura et al. 2015; Janka et al. 2016; Bruenn et al. 2016b; Burrows et al. 2018, 2020; Vartanyan & Burrows 2023; Sandoval et al. 2021; Burrows et al. 2024, 2026), uncertainties remain in which progenitors explode, what explosion energies they produce, and how these results vary across large progenitor grids (e.g. Takiwaki et al. 2014; Lentz et al. 2015; Melson et al. 2015a,b; Müller 2015; Janka et al. 2016; Hix et al. 2016). For a recent review of advances in multidimensional CCSN modelling, see Janka (2025).

In one-dimensional (1D), self-consistent, spherically symmetric simulations, CCSNe generally do not explode through the delayed neutrino-driven mechanism alone, except at the low-mass end of the progenitor population (e.g. Liebendörfer et al. 2004). Successful shock revival for more massive progenitors appears to require multidimensional hydrodynamic effects, particularly neutrino-driven convection and accretion shock instability, which increase the efficiency of neutrino heating behind the stalled shock (e.g. Herant et al. 1994; Janka & Mueller 1996; Blondin et al. 2003; Foglizzo et al. 2007). Alternatively, rapid rotation and magnetic-field amplification may produce magnetorotational explosions in a subset of strongly magnetised progenitors (Winteler et al. 2012; Mösta et al. 2014, 2015; Nishimura et al. 2015, 2017; Halevi & Mösta 2018; Kuroda et al. 2020; Obergaulinger & Aloy 2020; Aloy & Obergaulinger 2021).

As multidimensional simulations remain computationally expensive, several 1D approaches have been developed to parametrise the effects that facilitate successful explosions and apply them systematically across large progenitor grids. Simple prescriptions such as piston and thermal-bomb explosions, in which the shock is launched by moving an inner mass shell or by depositing thermal energy into the innermost stellar layers, have been widely used for nucleosynthesis calculations (e.g. Aufderheide et al. 1991; Woosley & Weaver 1995; Sawada & Maeda 2019). These approaches remain in use in recent studies (e.g., Farmer et al. 2023), and provide an efficient way to calculate nucleosynthetic yields across large progenitor grids. However, key explosion properties (i.e., the explosion energy and mass cut) are imposed rather than emerging from the explosion dynamics. More physically motivated 1D approaches attempt to approximate the multidimensional effects that facilitate shock revival. The Supernova Turbulence In Reduced-dimensionality (STIR) model incorporates neutrino-driven convection and turbulence through a time-dependent mixing-length prescription (Couch et al. 2020; Boccioli et al. 2021; Boccioli & Roberti 2026). The PUSH method, which is used in this study, instead introduces an additional source of neutrino heating to facilitate shock revival while retaining detailed neutrino transport (e.g., Perego et al. 2015; Ebinger et al. 2019). These approaches pro-

vide a framework for systematically studying explosion outcomes, remnant properties, and nucleosynthetic yields across large progenitor grids.

Assessing the impact of different stellar-evolution pathways—including binary stripping—requires considering not only how they alter the explodability of the resulting progenitors, but also how they affect the composition of the ejecta and the resulting observables (e.g. Nomoto et al. 2013; Bruenn et al. 2016a; Caffau et al. 2013). Nucleosynthetic yields therefore provide an important link between the pre-supernova evolution, the explosion dynamics, and the final composition of the ejecta. These yields depend on both the pre-supernova composition and stratification of the progenitor and on which regions ultimately become unbound during the explosion. In the outer ejecta, the final abundances are largely inherited from hydrostatic burning during stellar evolution, with the explosion primarily determining whether these layers are successfully ejected. This material contains the products of H, He, C, Ne, and O burning, including He, C, O, Ne, Mg, and other intermediate-mass elements. In contrast, the composition of the innermost ejecta is considerably more sensitive to the explosion itself, with the production of iron-group elements depending on the shock strength, the thermodynamic histories of the ejecta, the location of the mass cut, and the electron fraction, Y_e (see Curtis et al. 2019). Binary stripping can therefore influence the nucleosynthetic yields both by altering the pre-collapse composition and stratification of the progenitor and, through these structural differences, by affecting the conditions of the subsequent explosion.

The PUSH framework is particularly well suited to investigating how differences in progenitor structure affect nucleosynthetic yields. In this approach, electron-flavour neutrino luminosities and charged-current reactions are left unchanged, so the electron fraction of the innermost ejecta evolves self-consistently, while the mass cut emerges from hydrodynamic evolution rather than being imposed (Perego et al. 2015). Both aspects are important for predicting neutron-rich and proton-rich isotopes, particularly within the iron group (e.g. Fröhlich et al. 2006a,b). In contrast, piston-, thermal-, and kinetic-bomb approaches define the explosion energy and mass cut, limiting their ability to predict the nucleosynthesis of the innermost ejecta (e.g. Thielemann et al. 1996; Woosley & Weaver 1995; Rauscher et al. 2002; Limongi & Chieffi 2006; Woosley & Heger 2007; Umeda & Nomoto 2008; Limongi & Chieffi 2012).

In this work, we use the PUSH framework to simulate the single-star and binary-stripped progenitors of Farmer et al. (2023), allowing us to compare their explosion properties and nucleosynthetic yields consistently. We investigate how binary stripping alters the explodability of the progenitors, the resulting explosion energies and remnant masses, and the composition of the ejecta. The paper is structured as follows. In Section 2, we describe the progenitor models, the PUSH framework, and the nucleosynthesis post-processing. In Section 3, we show the explosion outcomes and properties across the progenitor grid and examine their dependence on ZAMS mass, compactness, and the adopted PUSH calibration. In Section 4, we explore the nucleosynthetic yields and the differences between the single-star and binary-stripped models. In Section 5, we compare our predictions with observational constraints from CCSNe and metal-poor stars. Finally, we summarise our main findings and conclusions in Section 6.

2 METHOD AND INPUTS

2.1 Input Models

In this study, we consider the two progenitor sets generated by [Farmer et al. \(2023\)](#): isolated single stars (the “Ms” series) and binary-stripped stars that have undergone Roche-lobe overflow (the “Mb” series). Both sets were calculated with the MESA stellar evolution code ([Paxton et al. 2011, 2013, 2015, 2018, 2019](#); [Jermyn et al. 2023](#)). Importantly, the Mb progenitors differ from their single-star counterparts not only in their envelope masses, but also in their core masses, composition gradients, and subsequent shell-burning evolution.

All progenitors are non-rotating and assume a solar metallicity of $Z = 0.0142$. The single-star and binary-stripped models adopt an initial helium mass fraction of $Y = 0.2684$ ($Y = 2Z + 0.24$; [Pols et al. 1995](#); [Tout et al. 1996](#)), with the initial abundance pattern based on the solar composition of [Grevesse & Sauval \(1998\)](#). The progenitor grids span ZAMS masses of 11–45 $M_{\odot}$ in intervals of 1 $M_{\odot}$. Four Mb models, with ZAMS masses of 24, 25, 28, and 29, are not evolved to the point of core-collapse within MESA due to numerical issues and are therefore excluded from our analysis. Each progenitor is identified by its ZAMS mass and series designation, Ms or Mb. Throughout this work, we place our results in the context of previous solar-metallicity PUSH calculations using the S-series ([Woosley et al. 2002](#)) and W-series ([Woosley & Heger 2007](#)) progenitors presented by [Ebinger et al. \(2019\)](#); [Curtis et al. \(2019\)](#) which are specified as WHW02 and WH07 throughout. Note that these single-star models were calculated from the KEPLER code, presented in [Woosley et al. \(2002\)](#) and [Woosley & Heger \(2007\)](#).

Following previous studies, we characterise the pre-collapse core structure using the compactness parameter introduced by [O’Connor & Ott \(2011\)](#). For consistency with the explodability analysis of [Farmer et al. \(2023\)](#), we evaluate the compactness at an enclosed mass of 2.5 $M_{\odot}$:

$$\xi_{2.5} = \frac{2.5}{R(M = 2.5 M_{\odot})/1000 \text{ km}}, \quad (1)$$

where $R(M = 2.5, M_{\odot})$ is the radius enclosing a baryonic mass of 2.5 $M_{\odot}$. Figure 1 shows $\xi_{2.5}$ as a function of ZAMS mass for the Ms, Mb, WHW02 and WH07 progenitor series. The compactness varies non-linearly with ZAMS mass, reflecting the sensitivity of the final core structure to shell-burning evolution, semiconvection, convective overshooting, and mass loss. Note that the PUSH calibration described below is based on the compactness at core bounce, $\xi_{2.0}$, following [Ebinger et al. \(2019\)](#). The corresponding $\xi_{2.0}$ trends with ZAMS mass are shown in Appendix A.

2.2 The PUSH Framework

We adopt the standard PUSH setup used in previous studies to follow the collapse, core bounce, shock revival, and the subsequent post-bounce evolution (see [Ebinger et al. 2019](#); [Curtis et al. 2019](#); [Ebinger et al. 2020b](#)). The resulting simulations provide the explosion properties, remnant masses, and thermodynamic trajectories required for nucleosynthesis post-processing. PUSH is based on the delayed neutrino-driven explosion mechanism and introduces an additional source of heating in the gain region of spherically symmetric simulations. This additional heating is motivated by the increased net heating experienced by fluid elements in multidimensional simulations, where convection and accretion can increase the efficiency of neutrino energy deposition. Specifically, PUSH deposits a fraction of the heavy-flavour neutrino energy ($\nu_x = \nu_{\mu}, \bar{\nu}_{\mu}, \nu_{\tau}, \bar{\nu}_{\tau}$) behind the shock to trigger successful explosions.

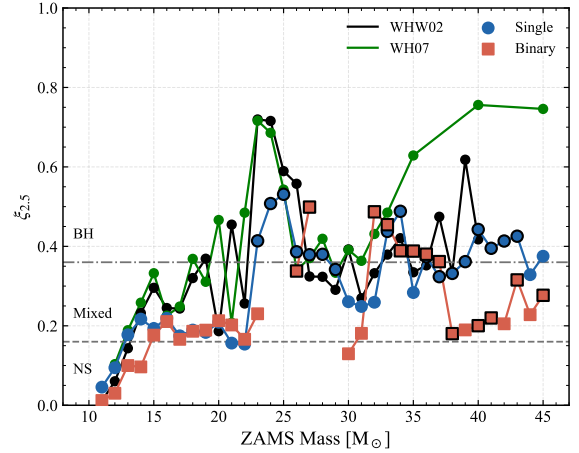


Figure 1. Compactness at 2.5 $M_{\odot}$ ($\xi_{2.5}$) across the progenitor ZAMS mass range for single and binary models. The horizontal thresholds at $\xi_{2.5} = 0.16$ and 0.36 indicate the approximate transition between neutron star (NS), mixed, and black hole (BH) outcomes ([Farmer et al. 2023](#)). Solar-metallicity progenitors from ([Woosley et al. 2002](#); [Woosley & Heger 2007](#)) are overplotted for comparison.

The method ([Perego et al. 2015](#)) employs the general-relativistic hydrodynamics code *Agile*, coupled to neutrino transport. Electron-flavour neutrinos are treated using the isotropic diffusion source approximation (IDSA; [Liebendörfer et al. 2001](#)), while heavy-flavour neutrinos are handled with the advanced spectral leakage (ASL; [Perego et al. 2015](#)) scheme. For matter in nuclear statistical equilibrium (NSE), we adopt the finite-temperature HS(DD2) equation of state ([Hempel & Schaffner-Bielich 2010a](#)). Outside NSE, the matter is described by an ideal-gas equation of state coupled to an α -network.

The PUSH heating term, $Q_{\text{push}}^+(t, r)$, transfers a fraction of the heavy-flavour neutrino energy to matter behind the shock and depends on the neutrino spectral energy flux and includes both spatial and temporal components, with two free parameters: k_{push} and t_{rise} . We adopt the standard calibration of [Ebinger et al. \(2019\)](#), based on SN 1987A, with $t_{\text{rise}} = 400$ ms and a compactness-dependent scaling

$$k_{\text{push}} = a\xi^2 + b\xi + c, \quad (2)$$

where $a = -23.99$, $b = 13.22$, and $c = 2.5$. Here, $\xi_{2.0}$ denotes the compactness at core bounce.

The simulations include material from the centre to the helium layer and are evolved for 15 s unless black-hole formation occurs earlier. The outcome of each model is classified as: (i) a successful explosion if the explosion energy saturates at a positive value, (ii) black-hole formation if the central density exceeds $\sim 10^{15} \text{ g cm}^{-3}$, or (iii) a failed explosion if the final explosion energy remains negative after PUSH becomes inactive. The explosion energy is computed following [Perego et al. \(2015\)](#) as the sum of thermal, kinetic, and gravitational energy integrated from the mass cut to the stellar surface.

2.3 Nucleosynthesis Post-processing

We perform detailed nucleosynthesis post-processing of successful explosion models using the nuclear reaction network CFNET [Fröhlich et al. \(2006b\)](#). Our setup follows [Ebinger et al. \(2019\)](#); [Curtis et al. \(2019\)](#), tracking 2,902 isotopes, including free nucleons and nuclei on both sides of the valley of β -stability up to ^{211}Eu . Reaction rates are

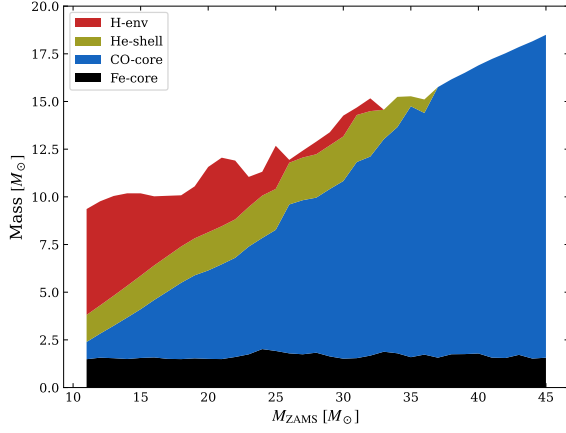


Figure 2. Fe-core mass (black), CO-core mass (blue), He-shell mass (yellow), and total mass including the H-envelope (red) as a function of ZAMS mass at the onset of collapse for the Ms series progenitors (Farmer et al. 2023).

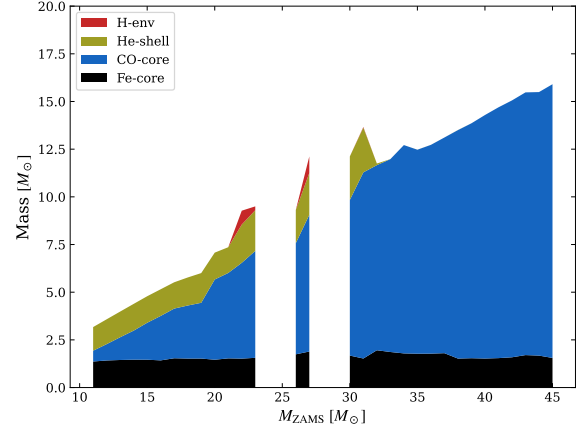


Figure 3. Same as Figure 2, but for the Mb series models (Farmer et al. 2023).

taken from REACLIB (Cyburt et al. 2010), incorporating experimental data where available and theoretical predictions for neutron-, proton-, and α -capture reactions from Rauscher & Thielemann (2000). Weak interactions include electron and positron capture rates from Langanke & Martínez-Pinedo (2001), $\beta^\pm$ decays from NuDat2¹ and Morrell et al. (2005), and neutrino and antineutrino captures on free nucleons.

The ejecta are divided into mass elements (“tracers”) of $10^{-3} M_\odot$. For each tracer, the thermodynamic history is followed from the onset of collapse to the final hydrodynamic timestep, up to 15 s, or until the shock leaves the computational grid. For tracers exceeding 10 GK, nucleosynthesis calculations begin once the temperature falls below this threshold, assuming NSE abundances as the initial composition. For cooler tracers, the calculations begin at the start of the hydrodynamic evolution, with the initial abundances taken from the nuclear reaction network coupled to the hydrodynamics. Note only tracers reaching temperatures above 1.75 GK are processed. In all cases, the electron fraction is initialised from the hydrodynamic simulation and evolved self-consistently within the network. The tracer trajectories are extrapolated beyond the simulation time by:

$$r(t) = r_{\text{final}} + t v_{\text{final}} \quad (3)$$

$$\rho(t) = \rho_{\text{final}} \left(\frac{t}{t_{\text{final}}} \right)^{-3} \quad (4)$$

$$T(t) = T[s_{\text{final}}, \rho(t), Y_e(t)] \quad (5)$$

where r , v , ρ , T , s , and Y_e denote radius, velocity, density, temperature, entropy per baryon, and electron fraction, respectively. The subscript “final” refers to values at the end of the hydrodynamic simulation. The temperature evolution is obtained using the equation of state of Timmes & Swesty (2000). Nucleosynthesis calculations are terminated once the temperature drops below 0.05 GK.

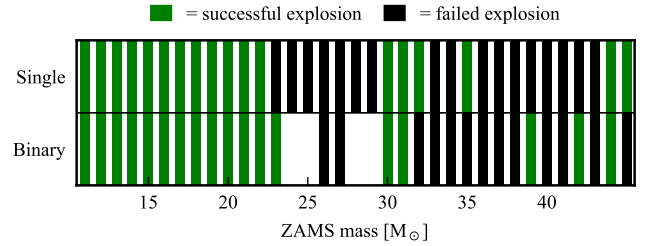


Figure 4. Explosion outcomes as a function of ZAMS mass for the Ms (top) and Mb (bottom) progenitor series, showing successful and failed models; gaps indicate masses for which no Mb model is available.

3 EXPLOSION PROPERTIES ACROSS THE MASS RANGE

In this section, we present and discuss the explosion outcomes and properties of our calculations for the full Ms and Mb progenitor sets. These models span ZAMS masses of 11–45 $M_\odot$, although four Mb models (24, 25, 28, and 29 $M_\odot$) are not evolved to the point of core-collapse within MESA and are therefore absent from our explosion calculations. We first discuss the successful and failed explosion outcomes across the mass range, before examining the resulting explosion energies, remnant masses, and explosion times as functions of ZAMS mass and $\xi_{2.0}$ compactness at bounce.

3.1 Explosion outcomes

Figure 4 summarises the explosion outcomes for the two progenitor series across the full ZAMS-mass range. We find that of the 35 Ms models investigated here 18 explode, corresponding to a success rate of 51.4 per cent, while 18 of the 31 Mb models explode, giving a slightly higher success rate of 58.1 per cent. The two series behave similarly at low ZAMS masses, with all models up to and including 22 $M_\odot$ exploding. Above this mass, however, their outcomes begin to diverge: the first Ms failure occurs at 23 $M_\odot$, whereas the first Mb failure occurs at 26 $M_\odot$. The two series produce opposite outcomes at 23 $M_\odot$, 32 $M_\odot$, 35 $M_\odot$, 42 $M_\odot$, and 45 $M_\odot$, while both explode at 30 $M_\odot$, 31 $M_\odot$, and 44 $M_\odot$.

Figure 5 compares our results with those of Farmer et al. (2023),

¹ <http://www.nndc.bnl.gov/nudat2/>

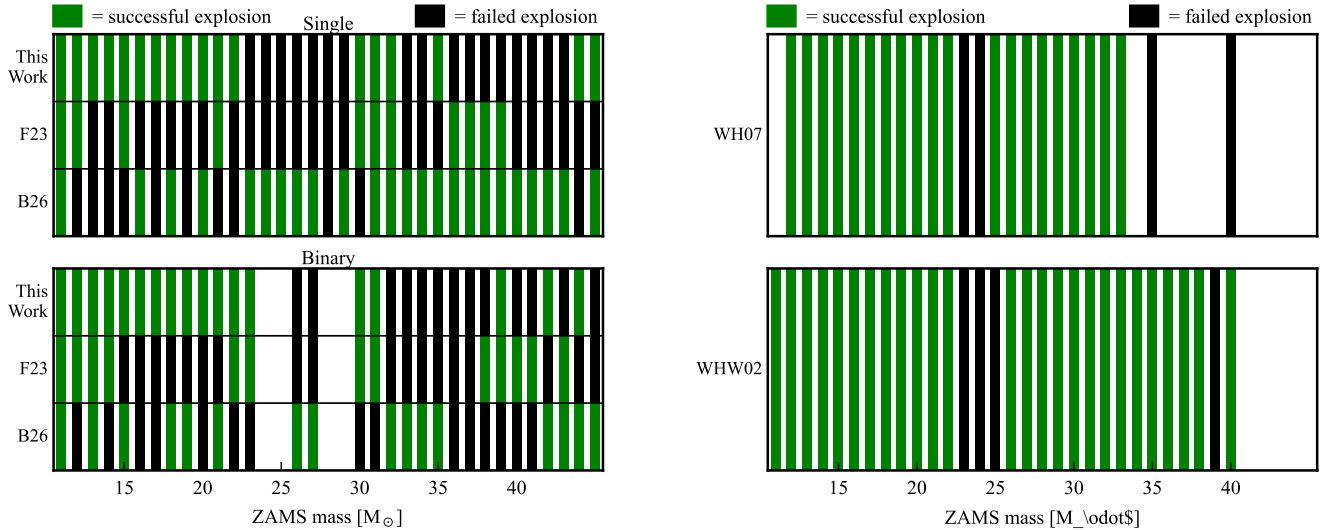


Figure 5. Left: Explosion outcomes as a function of ZAMS mass for the single-star (Ms; top) and binary-stripped (Mb; bottom) progenitor series. The outcomes obtained in this work are shown in the upper section of each panel. For comparison, the middle section shows the outcomes of Farmer et al. (2023), determined using the two-parameter explodability criterion of Ertl et al. (2016), while the lower section shows those of Boccioli & Roberti (2026), calculated using neutrino-driven explosion simulations with time-dependent mixing-length theory. Right: Explosion outcomes for the Woosley & Heger (2007) (top) and Woosley et al. (2002) (bottom) progenitor models from Ebinger et al. (2019) shown at the same integer ZAMS masses. Gaps indicate masses for which no model is available.

using the two-parameter explodability criterion of Ertl et al. (2016), and Boccioli & Roberti (2026), using neutrino-driven explosion simulations with time-dependent mixing-length theory. The three approaches disagree across substantial portions of the explodability landscape. For the Ms series at $M_{\text{ZAMS}} \leq 22 M_{\odot}$, all of our models explode, while Farmer et al. (2023) predict explosions only at 11, 12, 15, and $21 M_{\odot}$, and Boccioli & Roberti (2026) at 11, 16, 18, and $20 M_{\odot}$. Between 23 and $29 M_{\odot}$, none of our Ms models explode, in agreement with Farmer et al. (2023), whereas Boccioli & Roberti (2026) predict successful explosions for all but one model. At higher ZAMS masses, Boccioli & Roberti (2026) again predict explosions for most models, while our results show more frequent failures and somewhat better agreement with Farmer et al. (2023). A similarly mixed picture is found for the Mb series. All of our models up to and including $23 M_{\odot}$ explode, whereas both Farmer et al. (2023) and Boccioli & Roberti (2026) predict a mixture of successful and failed explosions and frequently disagree over individual models. Between 26 and $37 M_{\odot}$, our results are broadly consistent with Farmer et al. (2023), with both producing a mixture of outcomes. At the highest ZAMS masses, the three approaches again diverge. Overall, the limited agreement across both progenitor series highlights the strong sensitivity of the predicted explodability landscape to the adopted explosion treatment.

Figure 5 also shows the corresponding explosion outcomes for the Woosley et al. (2002) and Woosley & Heger (2007) progenitor sets, WHW02 and WH07. These models provide a useful comparison as they were used in the original PUSH calibration and nucleosynthesis studies (Ebinger et al. 2019; Curtis et al. 2019), but were evolved using different stellar-evolution assumptions from the Farmer et al. (2023) models explored here. Since the same explosion prescription is applied, differences between Figures 4 and 5 primarily reflect variations in the pre-collapse progenitor structures produced by the underlying stellar-evolution calculations. At low ZAMS masses, the WHW02 and WH07 progenitors show broadly similar behaviour to the Ms and Mb series, with all models up to and including $22 M_{\odot}$

exploding. The first failed explosion in both the WHW02 and WH07 sets occurs at $23 M_{\odot}$, matching the Ms series. The clearest difference between the progenitor grids appears between $25 M_{\odot}$ and $30 M_{\odot}$, where the WHW02 and WH07 sets produce more successful explosions than the Ms and Mb series. This behaviour is likely related to differences in the compactness distributions of the progenitor grids. In the standard PUSH calibration, k_{push} depends on the bounce compactness $\xi_{2.0}$ and reaches its largest values at intermediate compactness, around $\xi_{2.0} \approx 0.3$ (see figures 2 and 8 of Ebinger et al. 2019). Progenitor grids containing compactness troughs, or a larger number of models with intermediate compactness in this ZAMS-mass range, therefore receive stronger PUSH heating, allowing a greater fraction of models to explode. The role of this calibration is discussed further in Section 3.4. Compactness nevertheless provides only a one-parameter characterisation of the pre-collapse core structure and cannot uniquely determine the properties of the supernova; models with similar values of $\xi_{2.0}$ may still have substantially different density profiles, shell structures, and accretion histories, leading to differences in their explodability, nucleosynthetic yields, and synthetic observables. These differences are discussed in Section 4.2, where two models with similar compactness are compared. The differing outcomes of the Ms and Woosley progenitor sets also demonstrate that variations introduced during pre-supernova evolution can substantially alter the predicted explodability. Accurately modelling the pre-collapse stellar structure is therefore essential when assessing explosion outcomes with parametrised explosion methods such as PUSH.

3.2 Explosion properties as a function of ZAMS mass

In Figure 6, we compare the remnant masses obtained for the Ms and Mb progenitor series in this work with those of Farmer et al. (2023) and Boccioli & Roberti (2026). Although all three report a baryonic remnant mass, this quantity is determined differently in each approach. In the PUSH simulations, the proto-neutron star is

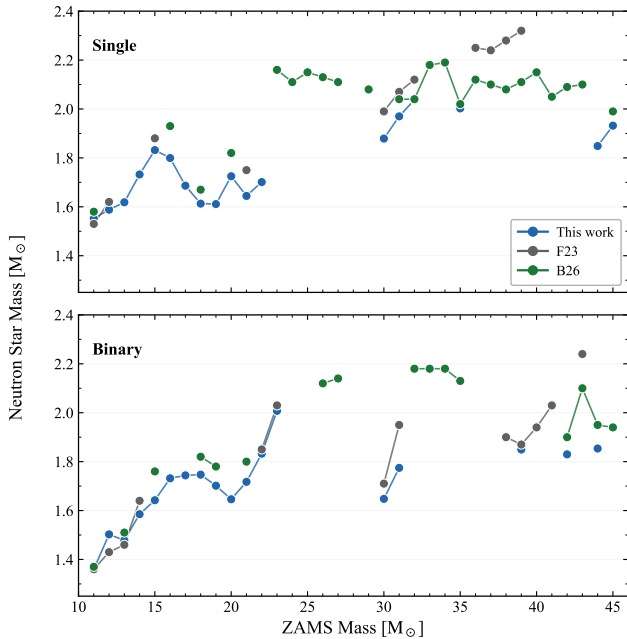


Figure 6. Remnant mass as a function of ZAMS mass for the successfully exploding single-star and binary-stripped models from this work (top), [Farmer et al. \(2023\)](#) (F23; middle), and [Boccioli & Roberti \(2026\)](#) (B26; bottom). Lines connect neighbouring successful models and are broken across gaps corresponding to failed explosions.

retained within the computational domain and the mass cut emerges self-consistently from the explosion dynamics. In contrast, [Farmer et al. \(2023\)](#) excise the material interior to the mass coordinate where the entropy per baryon reaches $s = 4$ before initiating the explosion and define the remnant mass as the compact-object mass at shock breakout, neglecting any fallback. In the neutrino-driven simulations of [Boccioli & Roberti \(2026\)](#), the mass cut also emerges self-consistently from the hydrodynamic evolution. The three methods also adopt different equations of state, with PUSH using HS(DD2) ([Hempel & Schaffner-Bielich 2010b](#)), [Boccioli & Roberti \(2026\)](#) using SFHo ([Steiner et al. 2013](#)), and [Farmer et al. \(2023\)](#) using the standard MESA EOS. These differences in numerical treatments, may therefore contribute to small systematic offsets between the calculations.

Despite the differences between the three methods, the predicted remnant masses are remarkably similar at low ZAMS masses when the same progenitor successfully explodes. Below $\sim 21\text{--}22 M_{\odot}$, the remnant masses generally agree to within $\sim 0.15 M_{\odot}$, with [Boccioli & Roberti \(2026\)](#) typically predicting the largest values, [Farmer et al. \(2023\)](#) intermediate values, and PUSH the smallest. At higher ZAMS masses, the predictions diverge more strongly. Where direct comparisons between successful Ms models are possible, PUSH consistently produces less massive remnants than [Farmer et al. \(2023\)](#) and [Boccioli & Roberti \(2026\)](#). The same trend is found for the Mb series, although the differences become particularly large above $\sim 40 M_{\odot}$.

Figure 8 shows the explosion energy, remnant mass, and explosion time of the successfully exploding Ms and Mb models as a function of ZAMS mass, with the compactness of each model indicated by the colour bar. The successfully exploding WHW02 and WH07 models from [Ebinger et al. \(2019\)](#) are included for comparison. Overall, the Ms and Mb models occupy broadly similar regions of parameter space, with explosion energies of $\sim 0.65\text{--}1.3 B$, remnant masses of

$\sim 1.5\text{--}2.0 M_{\odot}$, and explosion times of $\sim 0.35\text{--}0.65$ s. At low ZAMS masses, the Ms and Mb series produce similar remnant masses and explosion times. The main difference occurs between 11 and $14 M_{\odot}$, where the Mb models produce slightly lower explosion energies. These models also have lower compactness values, corresponding to lower values of k_{push} and therefore weaker additional heating. Between 15 and $22 M_{\odot}$, the two series are very similar in explosion energy, remnant mass, and explosion time. Larger differences emerge among the more massive exploding models, particularly above $\sim 30 M_{\odot}$, where the successful Ms models generally produce lower explosion energies, larger remnant masses, and later explosion times than the corresponding Mb models. The Ms models in this range also typically have lower compactness values. We note, however, that the pre-supernova evolution of massive stars is more uncertain in this mass range (e.g. [Hirschi 2015](#)), and the outcome of core collapse can be sensitive to the details of the adopted stellar-evolution prescriptions (e.g. [Sukhbold et al. 2016](#)).

The explosion properties of the Ms and Mb models are broadly consistent with those obtained previously for the WHW02 and WH07 progenitor sets. The clearest difference is in the explosion time, with the WHW02 and WH07 models generally exploding earlier. This likely reflects differences in both the underlying pre-collapse structures and in the exact evolutionary stage at which the progenitors are mapped into the hydrodynamic simulations. Finally at a fixed ZAMS mass, the differences between the single-star and binary-stripped progenitors can be comparable to the scatter among the different progenitor sets. We therefore examine the same quantities as a function of compactness in the following section.

3.3 Explosion properties as a function of compactness

Figure 9 shows the explosion energy, remnant mass, and explosion time as a function of the compactness at bounce, $\xi_{2,0}$. Compared with ZAMS mass, compactness reveals clearer trends across all three explosion properties. The strongest is found for the remnant mass: models with larger $\xi_{2,0}$ generally produce more massive neutron stars. This is expected as more compact progenitors sustain higher mass-accretion rates onto the PNS prior to shock revival, allowing the PNS to accrete more material before a successful explosion is launched ([Perego et al. 2015](#)).

The explosion energy also shows a broad dependence on compactness. In both series, the most energetic explosions occur at intermediate compactness, $\xi_{2,0} \sim 0.25\text{--}0.4$, while models at both lower and higher compactness generally produce less energetic explosions. This behaviour is closely related to the standard PUSH calibration, in which the compactness-dependent k_{push} reaches its largest values at intermediate $\xi_{2,0}$ and decreases towards both lower and higher compactness. The resulting energy trend therefore partly reflects the imposed compactness-dependent heating prescription, although substantial scatter remains at fixed compactness.

The explosion time shows a weaker dependence on compactness. Models with larger $\xi_{2,0}$ generally explode later than the lowest-compactness models, although there is considerable scatter within this relation. The remnant masses show a somewhat different behaviour: the scatter is largest at low and intermediate compactness, while above $\xi_{2,0} \sim 0.4$ the remnant masses become more tightly clustered across the different model series. This differs from the behaviour reported by [Boccioli & Roberti \(2026\)](#), who find substantially larger scatter in the mass cut of the high-compactness Ms progenitors from [Farmer et al. \(2023\)](#).

We note that in general the Ms and Mb models also show substantial overlap in compactness trends with the WHW02 and WH07

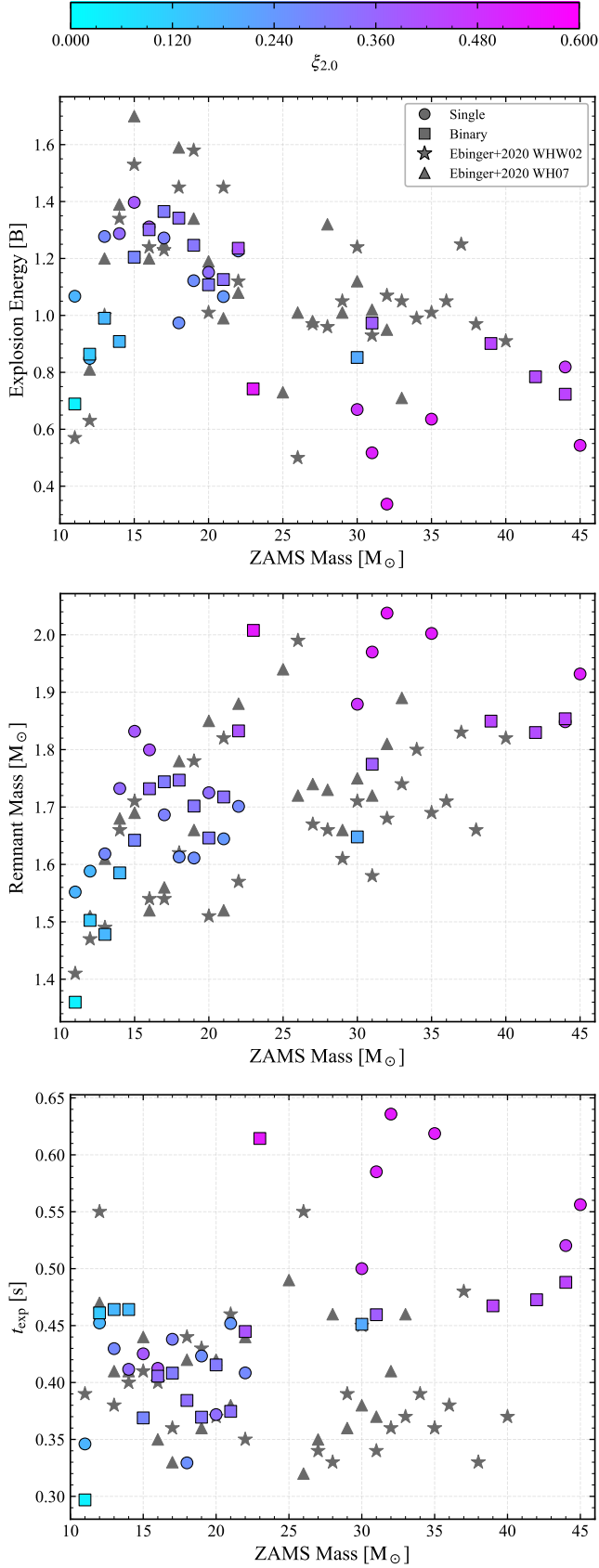


Figure 8. Explosion energy (top), neutron-star remnant mass (middle), and explosion time (bottom) as functions of ZAMS mass for the exploding Ms and Mb models. The WHW02 and WH07 models from Ebinger et al. (2019) are included for comparison. Symbol colour denotes $\xi_{2.0}$ at bounce.

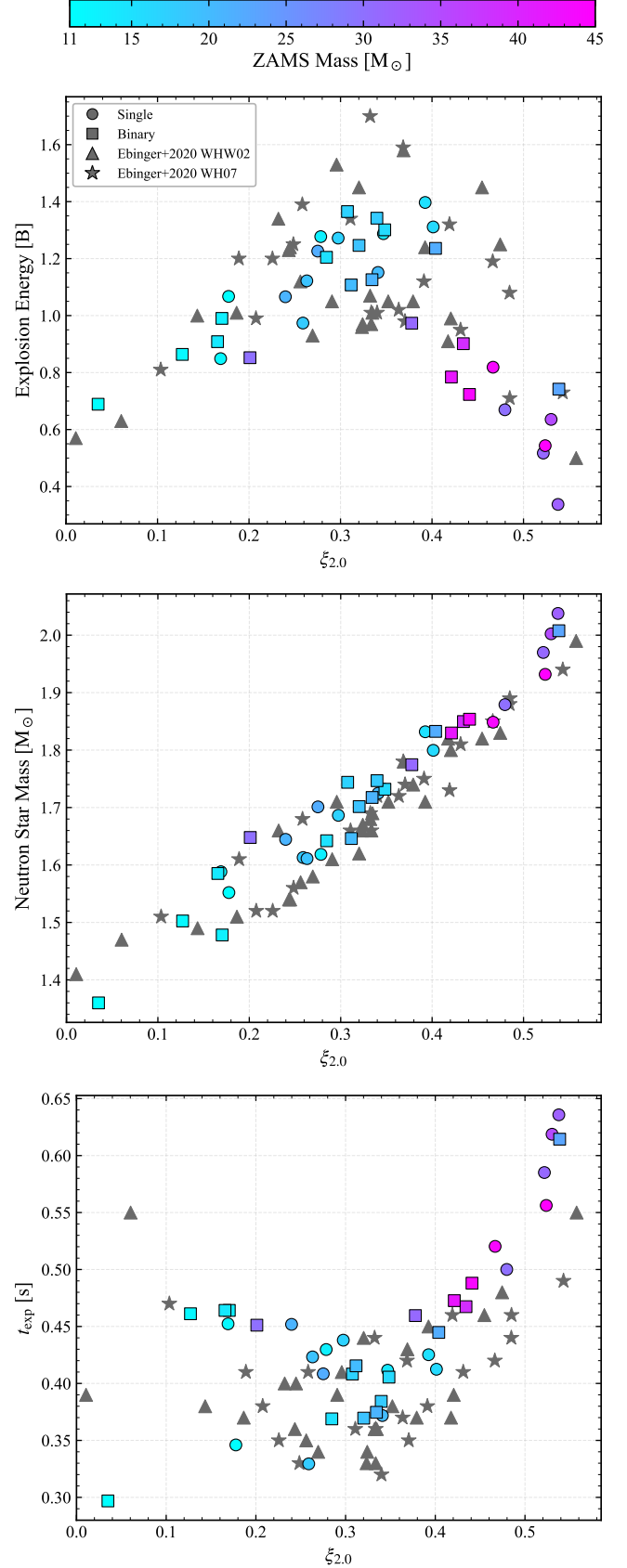


Figure 9. Explosion energy (top), neutron-star remnant mass (middle), and explosion time (bottom) as a function of $\xi_{2.0}$ at bounce, for the exploding Ms and Mb models. The WHW02 and WH07 models from Ebinger et al. (2019) are overplotted for completeness. Symbol colour denotes the ZAMS mass.

models, indicating that binary-stripped progenitors do not occupy a separate region of PUSH explosion parameter space. Compactness therefore provides a useful organising parameter for the explosion properties, but it does not capture all aspects of the pre-collapse structure. Models with similar $\xi_{2.0}$ can still differ in their density profiles, shell locations, and accretion histories, producing differences in both the timing and energetics of shock revival. In particular, [Boccioli & Roberti \(2026\)](#) find that the explodability of lower-compactness progenitors depends strongly on the density drop associated with the Si/O interface, whereas high-compactness explosions are less often triggered by accretion of this interface and instead arise from increasingly efficient neutrino heating. We also find that differences in the Si- and O-burning regions have an important impact on the resulting nucleosynthetic yields, which we discuss further in Section 4.

3.4 Sensitivity to the compactness-dependent k_{push} calibration

The comparisons above show that Ms and Mb progenitors with the same ZAMS mass can have different explosion outcomes. These differences cannot be interpreted from ZAMS mass alone as in the standard PUSH calibration, the strength of the additional heating is itself progenitor dependent. The parameter k_{push} is set by the compactness at bounce, so differences in compactness between an Ms and Mb progenitor affects the amount of additional PUSH heating they receive and therefore their hydrodynamic evolution.

To test how strongly the explosion outcome depends on this calibration, we considered the first pair with different outcomes, Ms23 and Mb23. Their bounce compactnesses are $\xi_{2.0} = 0.64$ and 0.54 , respectively. With the standard calibration, Ms23 fails to explode while Mb23 explodes successfully. We then repeated both calculations while keeping the progenitor structures unchanged but swapping the compactness values used to determine k_{push} . In other words, Ms23 was assigned the PUSH factor corresponding to Mb23, while Mb23 was assigned that of Ms23. The explosion outcomes reversed: Ms23 exploded with the Mb23 PUSH factor, whereas Mb23 failed with the Ms23 value. We performed the same test for the Ms32 and Mb32 pair, for which the standard calculations give the opposite outcome: Ms32 explodes while Mb32 fails. Swapping their PUSH factors again reversed the explosion outcomes. For both pairs examined here, the difference in outcome at fixed ZAMS mass is therefore strongly sensitive to the compactness.

These tests highlight an important caveat of the adopted parameterisation of k_{push} . For both model pairs considered here, changing only the PUSH factor is sufficient to reverse the explosion outcome. The predicted explodability therefore depends strongly on the assumption that compactness is the appropriate and dominant quantity with which to calibrate the additional heating. The differences between the Ms and Mb series must therefore be interpreted within the context of this calibration. Comparisons between progenitors with similar compactnesses are consequently particularly valuable, since they reduce differences arising directly from the compactness parameterisation and instead provide a clearer test of how the differences in progenitor structure affect the subsequent explosion properties and nucleosynthetic yields.

4 NUCLEOSYNTHESIS CALCULATIONS

We compute detailed nucleosynthetic yields for all successfully exploding models in the Ms and Mb series. In this section we focus on the Ms18.00 and Mb18.00 models, motivated in part by the association of an $\sim 18 M_{\odot}$ progenitor with SN 1987A. Their abridged

isotopic yields are presented in Appendix B, Table B1, while the full yield tables for all exploding models are provided electronically. Yields of radionuclides are also provided electronically in a similar format. We first briefly describe the dominant nucleosynthetic production channels from light to heavy elements in Section 4.1, before examining the Ms18.00 and Mb18.00 models in detail in Section 4.2. We then investigate how the nucleosynthetic yields vary across the full progenitor grid and with other model properties in Section 4.3, and examine production factors for selected models, in Section 4.4.

4.1 Overview of synthesis channels

The yields of light elements (H–N) are determined primarily by the pre-supernova evolution of the progenitor, through hydrostatic burning, convective mixing, and mass loss. Li, Be, and B are special cases, with significant contributions from other production sites and processes, including classical novae, low-mass stars, and cosmic-ray spallation. Neutrino interactions can also alter the abundances of rare isotopes in the outer ejecta through the ν -process ([Woosley et al. 1990](#); [Heger et al. 2005](#)), although these reactions are not included in our post-processing network. Since explosive burning has little effect on the bulk abundances of these lighter elements, their final ejecta reflect the composition and amount of material before core collapse. This is particularly important for the comparison between the Ms and Mb models, as binary stripping substantially alters the mass and composition of the outer stellar layers that are ultimately ejected.

The final yields of intermediate-mass elements (O–Ca) reflect contributions from both hydrostatic and explosive burning. During the explosion, additional intermediate-mass elements are synthesised primarily through explosive oxygen burning, although only a fraction of the progenitor's ^{16}O is processed in this way. Much of the O–Ne shell material does not reach temperatures high enough for substantial explosive burning. Consequently, elements such as O, Ne, and Mg retain substantial hydrostatic contributions, whereas the heavier IMEs are more sensitive to the explosion conditions. As a result, the yields of O, Ne, and Mg retain a strong hydrostatic contribution, while the heavier intermediate-mass elements are more sensitive to the thermodynamic conditions reached during the explosion. Their total yields therefore depend on both the progenitor structure and the explosion properties. Abundance ratios such as $[\alpha/\text{Fe}]$ provide a useful way to place these yields in the context of CCSN populations and galactic chemical evolution, where α denotes elements such as O, Ne, Mg, Si, S, Ar, Ca, and Ti.

The iron-group yields (Sc–Zn) originate predominantly from the deepest ejecta and are therefore sensitive to the explosion conditions. These species are synthesised mainly through complete and incomplete silicon burning, making their final yields dependent on the peak temperatures and densities reached by the innermost ejecta, as well as on the location of the mass cut. The electron fraction, Y_e , is especially important in these regions, as neutrino interactions can drive the ejecta away from the pre-supernova composition and strongly influence the relative production of neutron-rich and proton-rich isotopes. More neutron-rich conditions can favour isotopes such as ^{57}Ni relative to ^{56}Ni , while proton-rich material near the mass cut can enable limited νp -process nucleosynthesis and the production of nuclei beyond the iron group. The Fe-group and trans-iron yields therefore provide sensitive diagnostics of the thermodynamic and weak-interaction histories of the innermost ejecta, although their integrated yields may also contain contributions inherited from the pre-supernova composition.

Our nucleosynthesis post-processing adopts a mass resolution of $10^{-3} M_{\odot}$. To avoid drawing conclusions from the poorly resolved

late-time neutrino-driven wind, we exclude the three innermost tracer particles immediately next to the mass cut. We therefore do not attempt to draw conclusions from the detailed composition of the very innermost ejecta. In our calculations, material close to the mass cut reaches electron fractions as low as $Y_e \approx 0.35$. Despite these neutron-rich conditions, the ejecta do not reach the that required for a robust r -process (e.g., Farouqi et al. 2010). We also note that our calculations do not include neutrino-electron scattering, which contributes to neutrino thermalisation and spectral down-scattering and may consequently affect the electron fraction of the wind; see Fischer et al. (2012). For a more comprehensive discussion of the relevant caveats and nucleosynthetic production channels in PUSH calculations, see Section 3 of Curtis et al. (2019).

4.2 Two representative models: Ms18.00 and Mb18.00

The pre-explosion compositions of the inner layers of Ms18 and Mb18 are shown in the top panels of Figure 10, where the mass fractions of the dominant α -chain nuclei are plotted as a function of enclosed mass. Both models show a broadly similar stratification, with Si- and S-rich material immediately exterior to the iron core, followed by O- and Mg-rich layers at larger mass coordinates. The locations and extent of these regions differ between the two progenitors, with Mb18 showing more rapid transitions at the corresponding interfaces. The iron-rich region extends to $\sim 1.6 M_\odot$ in Ms18 and $\sim 1.7 M_\odot$ in Mb18, with the iron-group mass fraction falling below 10^{-3} beyond these points, whereas the Si-, S-, and Ar-rich layers extend farther outwards in Ms18, becoming subdominant to O- and Mg-rich material at $2.2 M_\odot$, compared with $\sim 1.9 M_\odot$ in Mb18. Moving farther outwards, the contrast between the progenitors becomes more pronounced. For clarity, Figure 10 only shows the inner regions of the two progenitors and does not extend through their full outer envelopes. However, in the Mb18 model the O-rich region declines sharply at $\sim 4 M_\odot$ and is followed by only a relatively small He-rich layer, with no H-rich envelope, whereas Ms18 retains a more extended O-rich region followed by substantial He- and H-rich outer layers. Binary stripping therefore modifies not only the total mass and compactness of the progenitors, but also the detailed stratification of the layers that ultimately contribute to the supernova ejecta.

We note that the PUSH simulations follow the collapse, core bounce, shock revival, and early shock propagation through the inner progenitor. The extended envelopes are therefore excluded from the hydrodynamic calculations and reattached afterwards (see Curtis et al. 2021). The resulting ejecta can then be evolved with radiation-hydrodynamics and Monte Carlo radiative-transfer codes, such as SNEC (Morozova et al. 2015) and SUPERNU (Wollaeger et al. 2013), to generate synthetic observables.

The bottom panels of Figure 10 show the post-processed compositions of the innermost ejecta exterior to the mass cut, together with the peak temperature and electron fraction reached by each layer. In both models, the innermost ejecta are dominated by iron-group isotopes, particularly ^{56}Ni , ^{57}Ni , and ^{58}Ni . The stable ^{58}Ni -rich material is concentrated interior to the ^{56}Ni -rich region, although its distribution extends to somewhat larger mass coordinates and higher mass fractions in Ms18 than in Mb18. Such differences could influence the synthetic observables, particularly the line profiles associated with stable and radioactive Ni, although detailed non-local thermodynamic equilibrium (NLTE) radiative-transfer calculations would be required to quantify this effect (e.g., Dessart 2025).

The mass cut lies within the Si/O-rich shell in both models, but at a larger enclosed mass in Mb18. This is consistent with previous PUSH investigations (e.g., Ebinger et al. 2019; Curtis et al. 2019)

and the trend seen in Section 3.3, where more compact progenitors generally produce larger remnant masses. At bounce, the two models have compactnesses of $\xi_{2.0} = 0.26$ for Ms18 and $\xi_{2.0} = 0.34$ for Mb18. The larger mass cut in Mb18 consequently leaves less of the innermost Si-rich material in the ejecta. More generally, the final iron-group yields depend on the amount of material exterior to the mass cut that is heated to temperatures sufficient for Si burning, while the peak temperature, density, expansion timescale, and electron fraction determine the balance between them and thus the resulting composition. The complete isotopic yields of both models are provided in Table B1; negative values in the explosive-yield columns indicate isotopes that undergo net destruction during the explosion.

Figure 11 shows the integrated final abundances of the layers that reach peak temperatures above 1.75 GK, corresponding to processed masses of $0.72 M_\odot$ and $0.68 M_\odot$ for Ms18 and Mb18, respectively. The two models exhibit broadly similar abundance patterns, with most isotopes produced at the same order of magnitude. The largest differences occur among the light and intermediate-mass nuclei. Differences in the lighter nuclei largely reflect variations in the initial composition and stratification of the material, while the scatter among intermediate-mass nuclei ($24 \lesssim A \lesssim 44$) reflects differences in the amount and distribution of material undergoing explosive O and incomplete Si burning. As shown in Figure 10, the initial ^{28}Si distributions differ between the two models. The Ms18 and Mb18 progenitors initially contain $0.294 M_\odot$ and $0.324 M_\odot$ of ^{28}Si , respectively, with the Si-rich material in Mb18 more strongly concentrated towards the inner layers. The larger mass cut in Mb18 therefore removes a greater fraction of this Si-rich material from the ejecta, while the material that is ejected also undergoes more extensive explosive processing. This results in the net destruction of $0.104 M_\odot$ of ^{28}Si in Ms18 and $0.178 M_\odot$ in Mb18, leaving final yields of $0.191 M_\odot$ and $0.146 M_\odot$, respectively. Thus, despite its larger initial ^{28}Si mass, Mb18 ejects less ^{28}Si as a greater fraction is processed during the explosion. This more extensive processing of the Si-rich material also contributes to differences among the heavier nuclei. In particular, the stronger abundance at $A = 56$ in the Mb18 model reflects its larger ^{56}Ni yield and is consistent with a greater amount of material undergoing complete Si burning.

4.3 Impact of Binary Stripping on Elemental Yields

In this section, we investigate the impact of binary stripping on the nucleosynthetic yields by comparing Ms and Mb progenitors at the same ZAMS mass. We first place our nucleosynthesis calculations in the context of previous studies. Figure 12 compares the yields obtained in this work with those of Farmer et al. (2023) and Boccioli & Roberti (2026). We focus on the Ms11 and Mb11 models, as $11 M_\odot$ is the only ZAMS mass for which both the single-star and binary-stripped models produce successful explosions in all three explosion treatments. The three calculations employ substantially different explosion and nucleosynthesis treatments. We post-process the thermodynamic trajectories with CFNET using a network of 2902 isotopes extending to ^{211}Eu , including neutrino and antineutrino captures. Boccioli & Roberti (2026) instead post-process their explosions with SKYNET using 1500 isotopes up to $A = 100$, without neutrino interactions during the post-processing. While Farmer et al. (2023) employs the substantially smaller 162-isotope MESA network up to ^{64}Zn .

Despite differences in approach, the three calculations recover broadly similar yield patterns, with most individual cases agreeing to within a factor of a few. The largest differences occur for

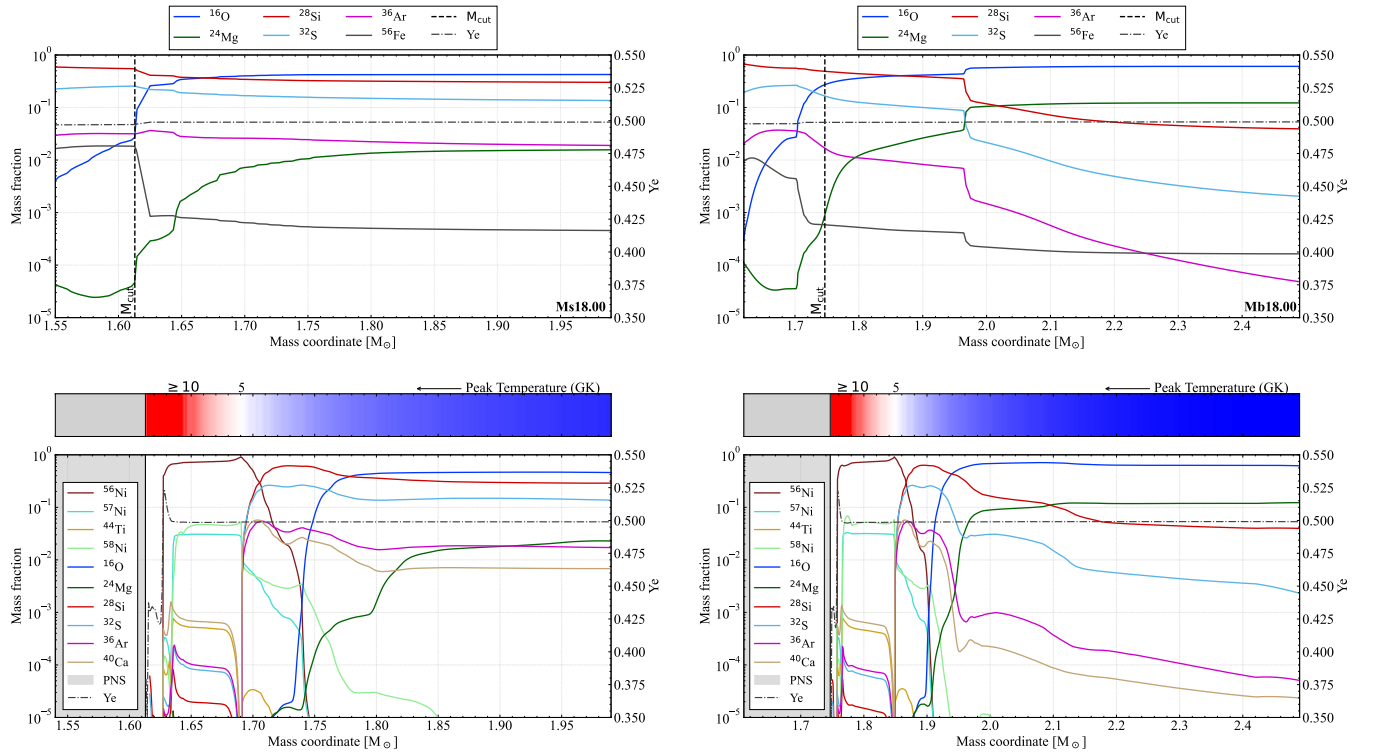


Figure 10. Top panels: pre-explosion composition of the inner layers of the Ms18.00 (left) and Mb18.00 (right) models. Bottom panels: final composition following explosive nucleosynthesis, with the colour scale indicating the peak temperature attained by each layer. The vertical dashed lines mark the mass cuts, while the dot-dashed curves show the corresponding electron-fraction profiles.

Table 1. Progenitor, explosion, and nucleosynthetic properties of the Ms18 and Mb18 models. The table columns give the ZAMS mass, $\xi_{2.0}$ at bounce, total progenitor mass at collapse, iron-core mass, carbon–oxygen core mass, helium-core mass, hydrogen-rich envelope mass, explosion energy, remnant mass, and the nucleosynthetic yields of selected isotopes. The iron-core mass is defined by the outer boundary of the central region with $Y_e < 0.495$. The CO-core mass is the enclosed mass at the outer boundary where the helium mass fraction satisfies $X_{\text{He}} \leq 0.2$. The helium-core mass is the enclosed mass at the outer boundary where the hydrogen mass fraction satisfies $X_{\text{H}} \leq 0.2$. The hydrogen-rich envelope mass is the mass beyond the helium core.

Model	M_{ZAMS} ($M_{\odot}$)	$\xi_{2.0}$ —	M_{prog} ($M_{\odot}$)	M_{Fe} ($M_{\odot}$)	M_{CO} ($M_{\odot}$)	M_{He} ($M_{\odot}$)	M_{env} ($M_{\odot}$)	E_{expl} (B)	M_{remn} ($M_{\odot}$)	^{56}Ni ($M_{\odot}$)	^{57}Ni ($M_{\odot}$)	^{58}Ni ($M_{\odot}$)	^{44}Ti ($M_{\odot}$)
Ms18	18	0.26	10.08	1.49	5.50	7.41	2.67	0.97	1.61	5.63E-02	1.82E-03	2.59E-03	2.86E-05
Mb18	18	0.34	5.74	1.52	4.31	5.74	0.00	1.34	1.75	7.78E-02	2.78E-03	4.48E-03	3.75E-05

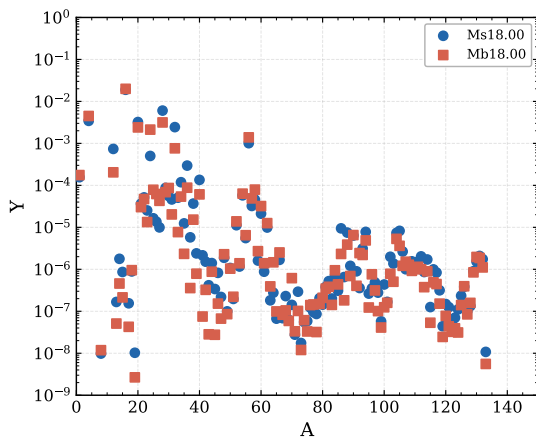


Figure 11. Mass-integrated final abundances Y as a function of mass number A for models Ms18.00 (blue circles) and Mb18.00 (red squares).

the iron-group elements, and most prominently for Zn. As noted by [Farmer et al. \(2023\)](#), Zn lies at the upper boundary of their reaction network, causing material to pile up in Zn isotopes that would otherwise be processed to heavier nuclei. More generally, differences among the iron-group yields are expected due to their sensitivity to the treatment of the innermost ejecta. [Boccioli & Roberti \(2026\)](#) compared previous PUSH calculations ([Curtis et al. 2019](#)) with their GR1D+ explosion and nucleosynthesis calculations and found that differences between the treatments can be traced to contributions from the νp -process, weak r -process elements, and the adopted NSE threshold. Furthermore, as iron-group elements are synthesised predominantly in the innermost regions of the ejecta through complete and incomplete Si burning, their yields are particularly sensitive to the explosion dynamics, mass cut, peak thermodynamic conditions, and local electron fraction. Therefore, when there is consensus when a given model should explode, the overall agreement between the three methods is remarkably good, with all methods showing that the binary stripping can produce substantial differences in the yields of particular elements.

Figure 13 shows the ratio of the final elemental yields from the binary-stripped progenitors to their corresponding single-star coun-

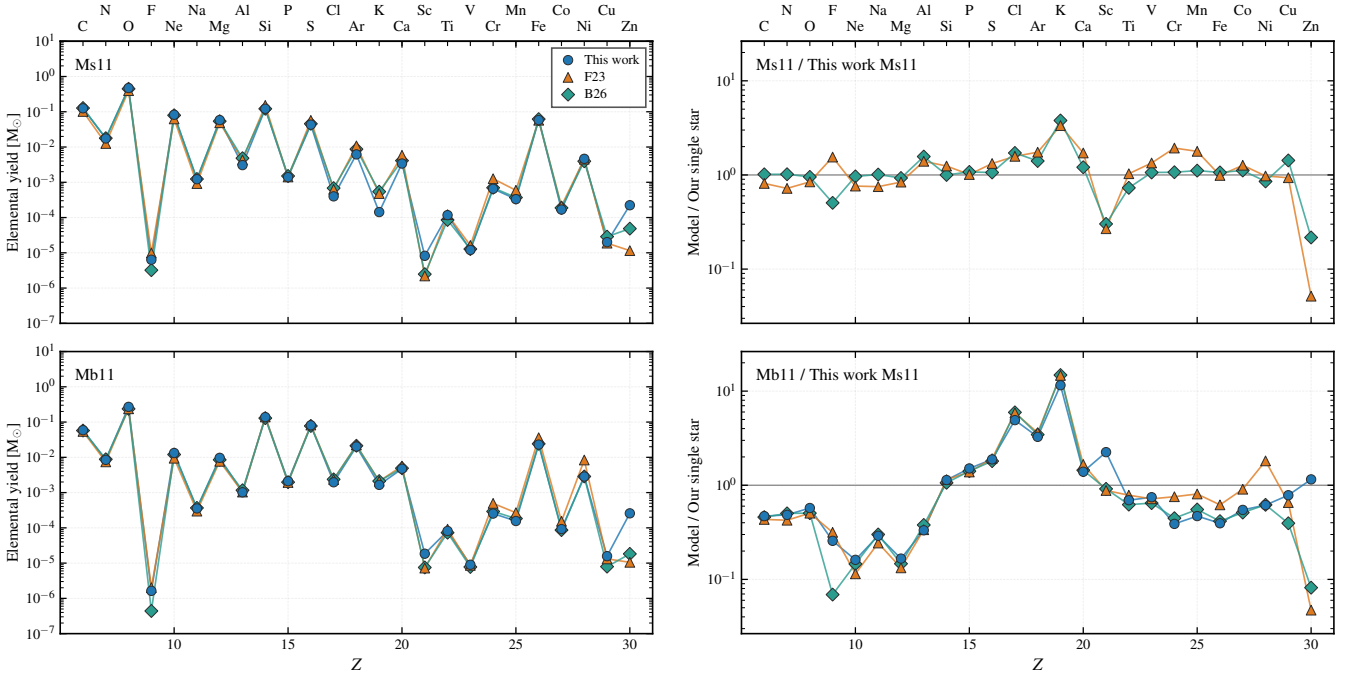


Figure 12. Elemental yields for the Ms11 (top) and Mb11 (bottom) models compared with those of [Farmer et al. \(2023\)](#) and [Boccioli & Roberti \(2026\)](#). The left-hand panels show the absolute elemental yields, whereas the right-hand panels show the corresponding yields relative to the Ms11 yields from this work (our single-star model). Note that in the upper-right panel only the [Farmer et al. \(2023\)](#) and [Boccioli & Roberti \(2026\)](#) models are shown and that the dashed horizontal line marks equal yields with the Ms11 model from this work.

terparts across the ZAMS-mass range. These final yields encompass both the pre-supernova evolution and the subsequent reprocessing of material during collapse and explosion. We find that the single-star progenitors typically have larger yields of lighter elements. This reflects the fact that these elements are predominantly produced during the hydrostatic evolution of the star and reside in the outermost regions. However, we find that binarity strongly affects the yields of heavier elements, with clear differences in the lowest (11–14 $M_{\odot}$) ZAMS-mass models between S–Sc. This is particularly interesting as S, Ar, and Ca are major products of explosive O burning and incomplete Si burning, while the production of Sc probes deeper and hotter layers, where it is largely produced through the decay of ^{45}Ti synthesized in high Y_e regions. Although the yields become more comparable at higher ZAMS masses. This reinforces an important result: binary stripping modifies not only the abundances of the lighter elements in the outer ejecta, but also the material that is subsequently processed during collapse and shock revival. This behaviour is consistent with [Farmer et al. \(2023\)](#), who found that binarity can substantially alter the final yields, with particularly strong effects for elements between Al and Sc. Similarly, [Boccioli & Roberti \(2026\)](#) found that the production of iron-group elements is especially sensitive to the details of the explosion method.

As the single-star and binary-stripped progenitor series do not explode at exactly the same corresponding ZAMS masses, it is difficult to isolate exact differences arising from the initial progenitor composition. We therefore remove the influence of the initial composition in Figure 14 by normalising each supernova yield by its corresponding pre-explosion abundance. We find that the lighter elements up to Al remain broadly similar, with the exception of the 44 $M_{\odot}$ progenitor, which shows a considerably larger relative enhancement. We find that this difference is driven by a small increase in the amount of N. Between Si and Sc, the single-star models typically show a greater

relative enhancement, whereas at intermediate ZAMS masses the binary-stripped models can show a somewhat greater relative enhancement of the heavier elements from Cr to Fe. The heaviest elements, which are most sensitive to the explosion treatment, generally show a slight relative enhancement in the binary-stripped models, with the clearest exception being Zn in the 13 $M_{\odot}$ progenitor. Zn is produced in complete Si burning and in material with $Y_e \gtrsim 0.5$ close to the mass cut. The inner structures of the binary-stripped progenitors differ substantially from their single-star counterparts, and in the Mb13 model the amount of Si-rich material close to the mass cut is approximately twice that in Ms13. This difference in the Si-rich material close to the mass cut thus partly drives the enhanced Zn production in Mb13 compared to the Ms13 model. However, it is difficult to infer exact trends as the compactness of the models is different and as such the PUSH heating term will also influence individual yields. We therefore examine the dependence of the nucleosynthetic yields on progenitor compactness in the following section for several important species.

Figure 15 shows, for each element at a given ZAMS mass, the ratio of its final abundance relative to Fe in the binary-stripped model to that in the corresponding single-star model. Normalising to Fe partly removes differences in the overall scale of the yields and highlights how the relative elemental abundance patterns vary between the two populations. We find no systematic enhancement relative to Fe across the full mass range, with the abundance patterns instead varying considerably with ZAMS mass. At the lowest ZAMS masses (i.e., 11, 13, 14 $M_{\odot}$) the binary-stripped models generally show enhanced intermediate mass element abundances, with several Fe-group elements also enhanced relative to the corresponding single-star models. Above this range, the binary-stripped models more commonly show reduced abundances relative to Fe, although considerable scatter remains, particularly among the intermediate-

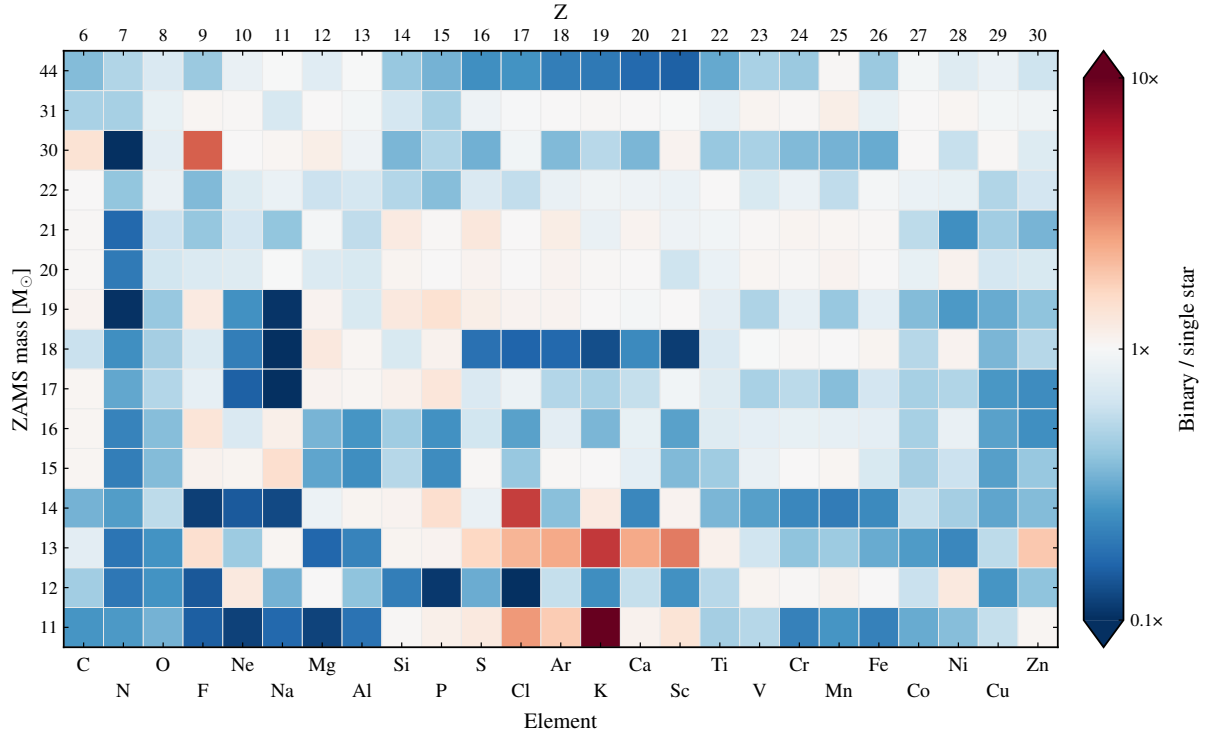


Figure 13. Ratio of the total elemental yields from the binary-stripped (Mb) progenitors to their corresponding single-star (Ms) progenitors as a function of ZAMS mass and element. Red indicates enhanced production in the binary-stripped model, while blue indicates enhanced production in the single-star model. Ratios are shown only for models for which both the Ms and Mb progenitors produce successful explosions.

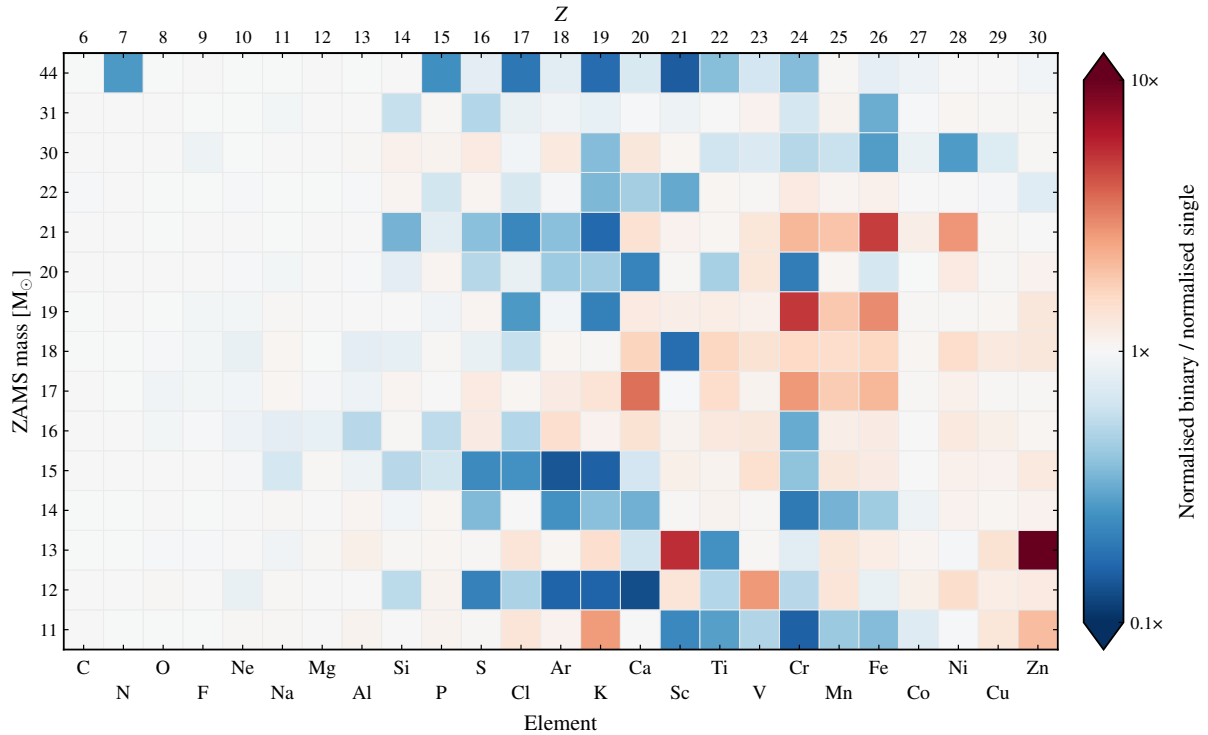


Figure 14. Similar to Figure 13, but with each elemental yield normalised by the corresponding initial progenitor abundance before taking the ratio between the binary-stripped and single-star models. Red therefore indicates greater relative enhancement in the binary-stripped model, while blue indicates greater relative enhancement in the single-star model.

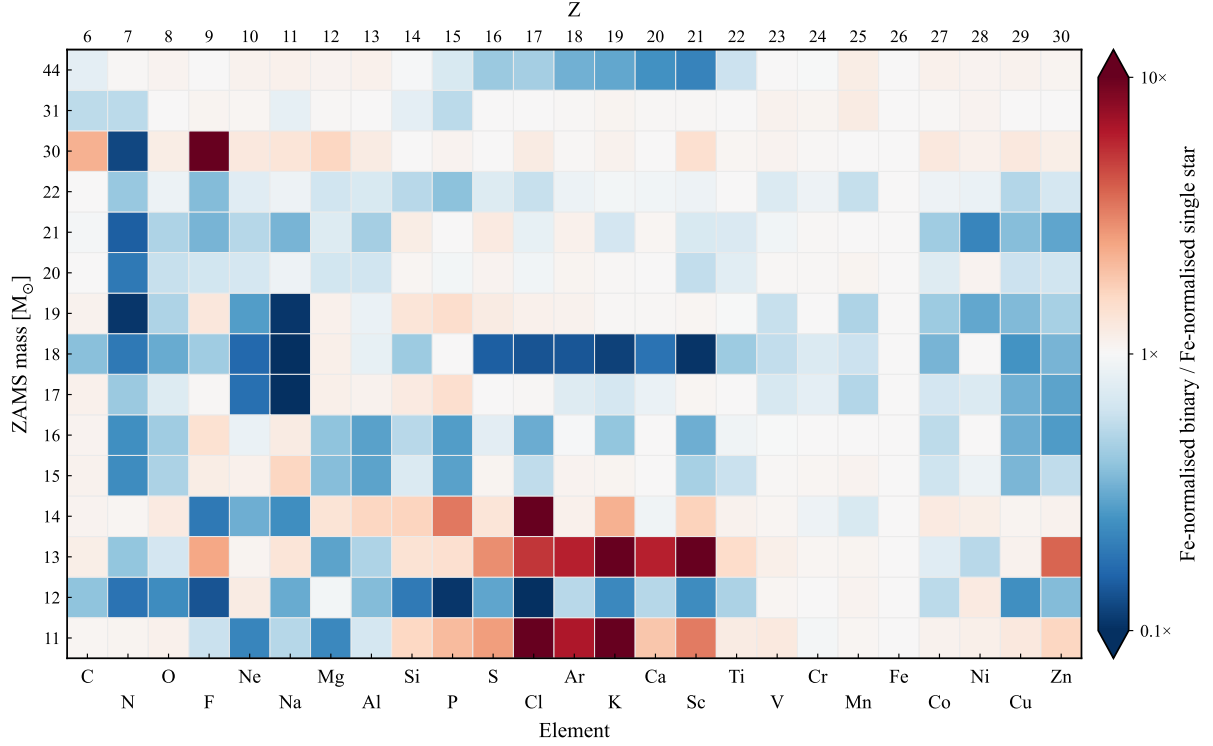


Figure 15. Ratio of the final elemental abundances relative to Fe in the binary-stripped (Mb) progenitors to those in their corresponding single-star (Ms) progenitors as a function of ZAMS mass and element. Red indicates a larger abundance relative to Fe in the binary-stripped model, while blue indicates a larger abundance relative to Fe in the single-star model. Ratios are shown only for models for which both the Ms and Mb progenitors produce successful explosions.

mass elements. Substantial variation is also present at the highest ZAMS masses (e.g., $44 M_{\odot}$), where several intermediate-mass elements are instead enhanced relative to Fe in the single-star models. This further demonstrates that binarity does not produce a uniform shift in the corresponding abundance pattern, with strong differences across the full ZAMS mass range and individual elements. Note we also compare several representative species with their corresponding solar abundance ratios in Section 5.3, providing further context for the predicted yields.

We conclude this subsection by assessing the population-level impact of binary stripping on the nucleosynthetic yields. We weight each progenitor by a Salpeter initial mass function (IMF; [Salpeter 1955](#)), $\phi(M) \propto M^{-2.35}$. The IMF weights are normalised over the available progenitor grid, and the IMF-weighted yield of each element is then calculated, where $Y_X(M)$ denotes the total yield of element X from a progenitor of ZAMS mass M . For progenitors that successfully explode, $Y_X(M)$ includes both the pre-supernova wind contribution and the final PUSH ejecta, while models that fail to explode contribute only their wind yields. Following [Boccioli & Roberti \(2026\)](#), the four binary-stripped progenitors that did not reach core collapse (24, 25, 28, and $29 M_{\odot}$) are excluded from the binary population. Since the IMF of binary-stripped stars is not known, we assume that this population also follows a Salpeter IMF.

Recently, [Rodríguez et al. \(2023\)](#) inferred a mean CCSN Fe yield of $0.058 \pm 0.007 M_{\odot}$, which [Weinberg et al. \(2024\)](#) subsequently used as an empirical anchor for the absolute scale of CCSN nucleosynthetic yields. Compared to this observational benchmark, our IMF-weighted Fe yield decreases from $0.0747 M_{\odot}$ for the single-star population to $0.0589 M_{\odot}$ for the binary-stripped population, bringing the latter into close agreement with the inferred mean. In general,

the integrated yields of the two populations remain broadly similar, although the combined effects of binary stripping, altered progenitor structure, and differences in explodability produce substantial changes for a small number of individual elements.

4.4 Yield trends with progenitor compactness and explosion properties

We now extend our analysis to the full set of successfully exploding Ms and Mb models to examine how the nucleosynthetic yields vary across the progenitor landscape. Following [Curtis et al. \(2019\)](#), we consider the yields of key isotopes and iron-group elements primarily as a function of progenitor compactness, while also examining differences associated with ZAMS mass and pre-explosion structure.

Figure 17 shows the yields of ^{56}Ni , ^{57}Ni , ^{58}Ni , and ^{44}Ti as a function of $\xi_{2.0}$, with the models colour-coded by ZAMS mass. This allows the yield trends to be considered alongside the progenitor structures shown in Figures 2 and 3. Several general trends are apparent. The yields of ^{56}Ni and ^{44}Ti increase with compactness, while ^{57}Ni follows a similar trend but with greater scatter. More compact progenitors tend to produce larger yields, although the strength of this relation varies between isotopes. At higher compactness, the previous PUSH calculations show somewhat less scatter than the single-star and binary-stripped models investigated here for ^{56}Ni , ^{57}Ni , and ^{44}Ti . The ^{58}Ni yields do not show the same behaviour and show a large scatter across the entire compactness range. This demonstrates that, at higher compactness, variations in nucleosynthetic yields resulting from binary evolution can be comparable to, or even greater than, those arising from differences in the underlying progenitor calculations, including the adopted nuclear reaction networks.

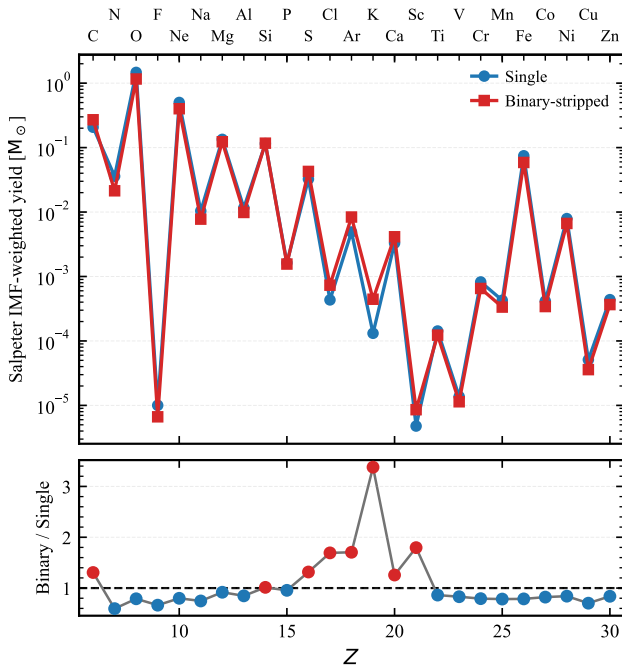


Figure 16. Salpeter IMF-weighted elemental yields for the single-star (Ms) and binary-stripped (Mb) progenitor populations, with the corresponding yield ratio shown in the lower panel. Red points in the lower panel indicate greater production by the binary-stripped population, whereas blue points indicate greater production by the single-star population. The dashed horizontal line marks equal production between the two populations.

The ZAMS-mass colour-coding provides further insight into this scatter, showing that models with similar compactness can originate from very different progenitor structures and therefore produce substantially different nucleosynthetic yields. For example, Ms35 and Ms45 both have $\xi_{2.0} \approx 0.52$, yet Ms45 produces almost twice as much ^{56}Ni . Inspection of the pre-explosion structures shows that Ms45 has a larger and more compressed Si-rich region available for explosive processing, allowing more material to undergo Si burning and contribute to the radioactive Ni yield. Some scatter is also present within the binary-stripped sequence, although the differences between models of comparable compactness are generally smaller than those found among the single-star models. This illustrates that compactness is an important predictor of the nucleosynthetic yields, but does not by itself capture all of the progenitor-structure differences that influence the final abundances.

The weakest relationship is found for the stable, neutron-rich isotope ^{58}Ni . Although the most compact models tend, on average, to produce somewhat larger ^{58}Ni yields, the models do not follow a single well-defined trend. This is broadly consistent with Curtis et al. (2019), who found that the production of neutron-rich isotopes depends on the local electron fraction, Y_e . Rather than following a single compactness-dependent relation, ^{58}Ni yields separate into multiple branches associated with different Y_e conditions, with lower Y_e generally favouring greater production of neutron-rich isotopes such as ^{57}Ni and ^{58}Ni .

Figure 18 shows the final elemental yields of the iron-group elements Ti, Cr, Mn, and Ni as a function of compactness, with the models colour-coded by ZAMS mass. These trends can be understood through the dominant isotopes and burning regimes responsible for their production. Ti and Cr are produced predominantly through

the symmetric nuclei ^{48}Cr and ^{52}Fe , which decay to ^{48}Ti and ^{52}Cr , respectively. Most Mn is produced as ^{55}Co during incomplete Si burning, while Ni is produced both directly as stable Ni isotopes and through the decay of nuclei such as ^{60}Zn and ^{60}Cu during complete Si burning (see Figure 4 of Curtis et al. 2019). Focusing on the highest-ZAMS-mass models, where the differences in the yield trends are most pronounced, these variations can be traced directly to differences in their pre-explosion structures. In Mb44, the Si-dominated region above the mass cut extends from $\sim 1.8 M_{\odot}$ to $\sim 2.3 M_{\odot}$, beyond which O-rich material becomes dominant, whereas in Ms44 the Si-rich region extends to $\sim 2.6 M_{\odot}$. Ms44 therefore has a larger reservoir of Si-rich material available for explosive processing, allowing more material to undergo complete Si burning and enhancing the production of Ti, Cr, and Ni. Mn follows this behaviour less closely, reflecting its stronger association with incomplete Si burning and its sensitivity to the local electron fraction.

5 COMPARISON WITH OBSERVATIONAL CONSTRAINTS

To place the predicted explodability and nucleosynthetic yields in an observational context, we compare the single-star and binary-stripped models with constraints from individual CCSNe in Section 5.1, abundance patterns in metal-poor stars in Section 5.2, and discuss the predicted α -element abundance ratios in Section 5.3.

5.1 Comparison with the observed landscape

Figure 19 shows the ejected ^{56}Ni mass as a function of ZAMS mass for the single-star and binary-stripped explosion models. We also include the results from the exploding models of Ebinger et al. (2019), calculated by Curtis et al. (2019), together with observationally inferred ^{56}Ni masses for CCSNe from Nomoto et al. (2013) and Bruenn et al. (2016a). These observational estimates are restricted to neutrino-driven CCSNe and exclude HNe. The observational sample spans progenitors with ZAMS masses of ~ 10 – $23 M_{\odot}$ and inferred ^{56}Ni masses of ~ 0.01 – $0.09 M_{\odot}$. There are no observational constraints for progenitors with ZAMS masses above $\sim 25 M_{\odot}$. Both the single-star and binary-stripped models broadly cover the observed range, up to $\sim 23 M_{\odot}$, the predicted ^{56}Ni masses vary relatively little, and the scatter is generally smaller than that found in the previous PUSH calculations for models up to $\sim 23 M_{\odot}$. At higher ZAMS masses, the scatter increases considerably. While Curtis et al. (2019) found that the ^{56}Ni yields remained relatively constant at high ZAMS masses, the models investigated here exhibit substantially greater variation. This is particularly evident for the Ms44 and Ms45 models, which produce approximately twice as much ^{56}Ni as the corresponding binary-stripped models. As discussed above, this difference is associated with variations in their pre-explosion structures and in the amount of Si-rich material available for explosive processing.

Another notable difference occurs between 11 and $15 M_{\odot}$, where the binary-stripped models generally produce weaker explosions and smaller ^{56}Ni yields. The Mb11 model, in particular, lies close to the two lowest observationally inferred ^{56}Ni masses, whereas the corresponding Ms11 model ejects considerably more ^{56}Ni . Since the ^{56}Ni yield scales approximately linearly with compactness, the low compactness of Mb11 naturally explains its lower yield relative to the single-star model at the same ZAMS mass. Nevertheless, the Mb11 yield remains consistent with the lower end of the previous PUSH calculations from Ebinger et al. (2019).

We also compare the predicted Ni/Fe ratios from the single-star

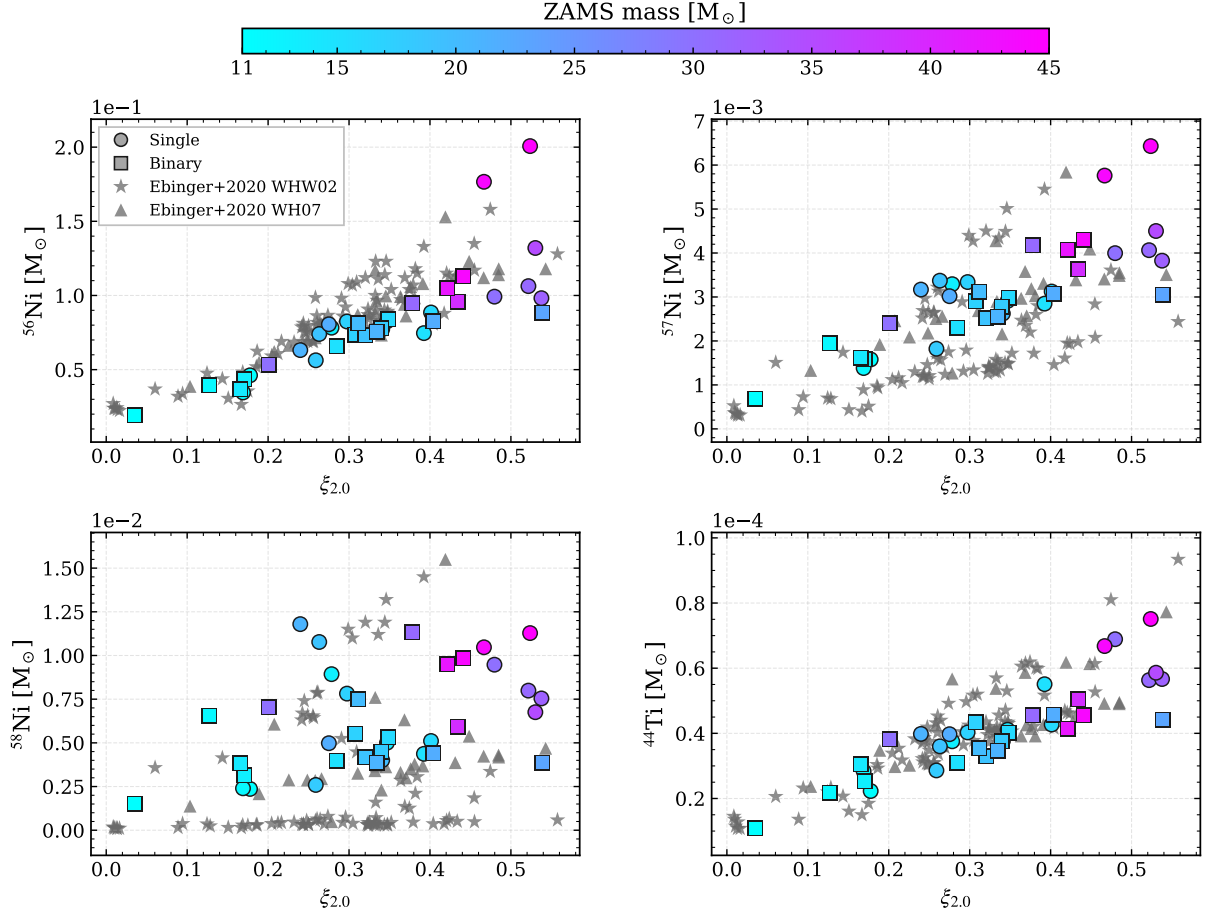


Figure 17. Isotopic yields of ^{56}Ni (top left), ^{57}Ni (top right), ^{58}Ni (bottom left), and ^{44}Ti (bottom right) after explosive processing, shown as a function of compactness for the single-star (blue circles) and binary-stripped (red squares) model series. The corresponding yields from [Curtis et al. \(2019\)](#) are overplotted for the WHW02 and WH07 model series as grey triangles and stars, respectively.

and binary-stripped models with the observationally inferred values compiled by [Jerkstrand et al. \(2015\)](#) for SN 1987A, SN 2004et, SN 2012aw, and SN 2012ec in Figure 20. The comparison is restricted to models with ZAMS masses between 12 and 21 $M_{\odot}$, with marker colour indicating the $\xi_{2.0}$ compactness at bounce and the dashed horizontal line marking the solar Ni/Fe ratio. The low-compactness Mb12 and Mb14 models provide the closest overall agreement with SN 2012ec, reproducing both its supersolar Ni/Fe ratio and its comparatively large ^{56}Ni yield. The Ms12 and Mb14 models produce similar amounts of ^{56}Ni , however, their predicted Ni/Fe ratios differ from those observed. Several of the higher-compactness models are consistent with the ^{56}Ni yields and Ni/Fe ratios inferred for SN 1987A and SN 2012aw. However, none of the models produce a sufficiently low Ni/Fe ratio to reproduce SN 2004et. Overall the predicted yields span much of the observed range for these low-mass explosions.

5.2 Comparison with metal poor stars

Massive stars enriched the early Universe through their winds and subsequent supernovae. Therefore, the atmospheric compositions of long-lived, low-mass, metal-poor stars can preserve the chemical signatures of the first generations of core-collapse supernovae, thereby providing observational constraints on predicted nucleosyn-

thetic yields. Numerous photometric and spectroscopic surveys have identified metal-poor stars and characterised their atmospheric abundances through spectral analyses (e.g. [Caffau et al. 2013](#); [Norris et al. 2013](#); [Keller et al. 2014](#); [Li et al. 2015a,b](#); [Howes et al. 2015](#); [Placco et al. 2015](#); [Jacobson et al. 2015](#); [Hansen et al. 2015](#); [Roederer et al. 2016](#); [Abomalima & Frebel 2018](#); [Starkenburg et al. 2017](#); [Andrievsky et al. 2018](#)). Moreover, since the publication of [Curtis et al. \(2019\)](#), several additional studies have expanded the search and characterisation of metal-poor stars (e.g. [Da Costa et al. 2019](#); [Marino et al. 2019](#); [Venn et al. 2020](#); [Yong et al. 2021](#); [Aoki et al. 2022](#); [Li et al. 2022](#); [Zepeda et al. 2022](#); [Arentsen et al. 2023](#); [Martin et al. 2024](#)). [Snedden et al. \(2016\)](#) used improved laboratory transition data for neutral and singly ionised iron-group species to derive detailed abundances for the very metal-poor main-sequence turnoff star HD 84937. In Figure 21, we compare these abundances with the elemental [X/Fe] ratios predicted by our single-star and binary-stripped models. The triangles show the abundances inferred from neutral and singly ionised species, with error bars corresponding to 3σ uncertainties, while the semi-transparent horizontal segments show the predictions of the individual progenitor models. Following [Curtis et al. \(2019\)](#), we show the individual model predictions rather than weighting the yields by an IMF to highlight the variation within the model sets.

Overall, the predicted iron-group abundance pattern is in good

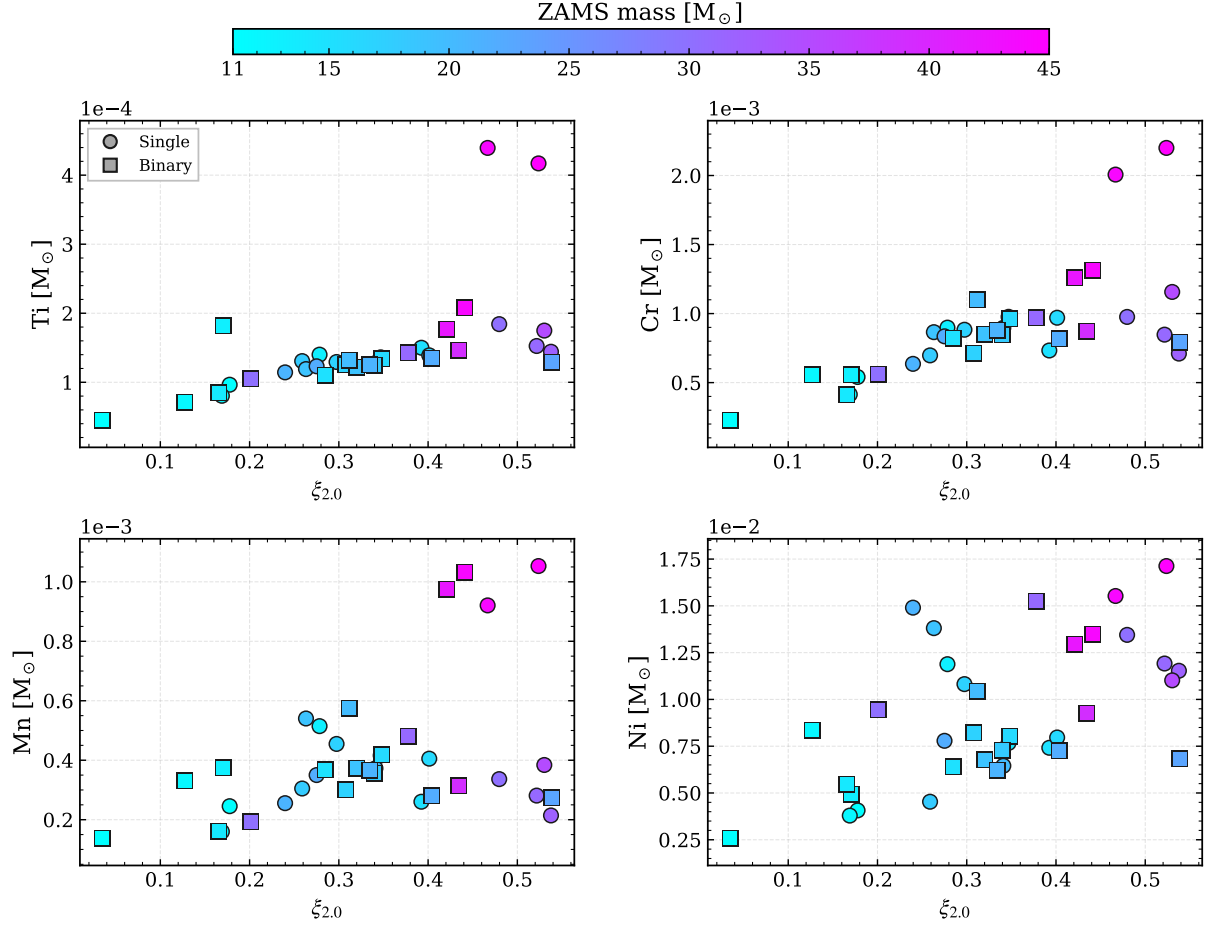


Figure 18. Elemental yields of titanium (top left), chromium (top right), manganese (bottom left), and nickel (bottom right) after explosive processing as a function of compactness for both single-star (blue circles) and binary-stripped (red squares) model series.

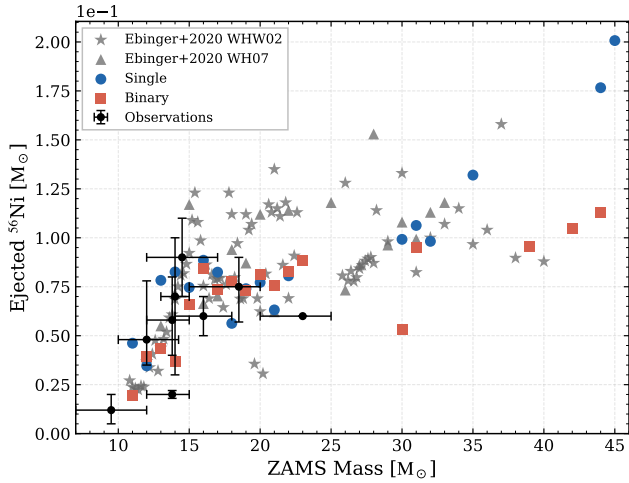


Figure 19. Yields of ^{56}Ni as a function of ZAMS mass for the single-star (blue circles) and binary-stripped (red squares) model series. Observational estimates for faint supernovae are overplotted as black crosses with error bars (see [Nomoto et al. 2013](#); [Bruenn et al. 2016a](#), for further details). The corresponding yields from [Curtis et al. \(2019\)](#) are overplotted for the WHW02 and WH07 model series as grey triangles and stars, respectively for comparison.

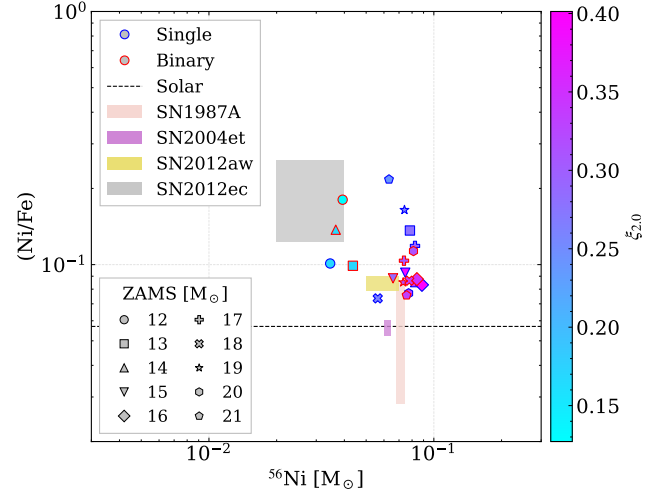


Figure 20. Ni/Fe ratio as a function of ejected ^{56}Ni for SN 1987A, SN 2004et, SN 2012aw, and SN 2012ec (see [Jerkstrand et al. 2011](#)). The observations are compared with the single-star and binary-stripped model series, restricted to the ZAMS mass range 12–21 $M_{\odot}$, with marker symbols denoting ZAMS mass. The dashed horizontal line marks the solar Ni/Fe ratio. No uncertainties in the Ni/Fe ratios were reported for SN 2004et and SN 2012aw; these points are therefore shown with error bars comparable to the marker size for visibility.

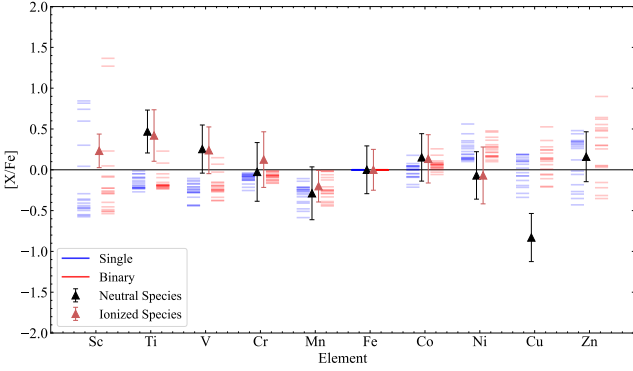


Figure 21. Elemental abundance ratios, $[X/Fe]$, for the single-star (blue) and binary-stripped (red) model series, compared to the observed neutral and ionized abundances of HD 84937 (black and red triangles, respectively), with 3σ uncertainties. Model predictions are shown as short horizontal segments, offset to the left (single-star) and right (binary-stripped) of each element for clarity, with each segment corresponding to an individual model. The transparency of the segments is normalized per series such that darker colours indicates a larger number of models predicting a particular abundance ratio, with all models in a series combined producing the full colours shown in the legend.

agreement with that observed in HD 84937, although substantial model-to-model variation is present for several elements. The spread between the single-star and binary-stripped models is comparable to the variation between the WHW02 and WH07 progenitors found by Curtis et al. (2019). The largest spread is seen for Sc and Zn, both of which have traditionally been challenging to reproduce in sufficient quantities in CCSN calculations. Nevertheless, subsets of both the single-star and binary-stripped models reproduce the observed Sc and Zn abundances. Their production is particularly sensitive to conditions in the innermost ejecta. The largest $[Sc/Fe]$ ratios are obtained for the Mb11 and Mb13 models, consistent with the enhanced Sc yields seen in Figure 13. Both of these models also show enhanced $[Zn/Fe]$ relative to most of the model set. We find that the clearest discrepancy with HD 84937 is found for $[Cu/Fe]$, which is systematically overproduced in our calculations. Cu production is sensitive to the neutron excess and the detailed thermodynamic and Y_e evolution of the innermost ejecta, and can therefore vary substantially between progenitor models and explosion treatments. Curtis et al. (2019) found Cu abundances consistent with the observations for their WHW02 progenitors, whereas their WH07 calculation did not, further illustrating this sensitivity. Finally we note that both model sets also produce a broad range of $[Mn/Fe]$ ratios, many of which lie within the observational constraints.

5.3 Alpha elements

The preceding sections have primarily focused on the Fe-group elements, we now investigate trends in alpha elements. We discuss three representative alpha nuclei: ^{16}O , ^{28}Si , and ^{40}Ca . In metal-poor stars the observed surface abundances largely reflect enrichment by the interstellar medium by earlier core-collapse supernovae, particularly when $[Fe/H] \lesssim -1$. At higher metallicities, the delayed contribution of Type Ia supernovae becomes increasingly important and lowers the characteristic alpha-to-iron ratio, as they produce a considerable amount of Fe-group elements. Metal-poor stars typically exhibit alpha enhancements of $\sim [\alpha/Fe] \approx +0.3$ – $+0.5$, although there is considerable scatter between individual elements and stellar

populations (Cayrel et al. 2004; Nomoto et al. 2013). It is therefore useful to determine whether CCSNe from binary-stripped and single-star progenitors produce systematically different alpha-to-iron abundance ratios. For an isotope AX , we calculate its abundance relative to Fe and normalise this ratio to the corresponding solar value. We adopt the solar elemental abundances of Asplund et al. (2009) and the isotopic fractions of Lodders et al. (2009).

It is expected that the clearest dependence on ZAMS mass will be for that of $[^{16}O/Fe]$. This is because most of the ejected ^{16}O is inherited from hydrostatic burning and is located in the O-rich layers of the progenitor. As the ZAMS mass and the CO-core mass increase, the amount of oxygen available to be ejected generally also increases. In contrast to this the final Fe abundance is dominated by the decay of explosively synthesised ^{56}Ni which depends strongly on the explosion properties and final mass cut. As shown in Figure 19, the ejected ^{56}Ni mass initially increases with ZAMS mass in our models but begins to plateau around $15 M_{\odot}$. The continued growth of the oxygen-rich layers, without a corresponding increase in the Fe yield, should therefore produce larger $[^{16}O/Fe]$ values for the more massive successful explosions, while departures from this trend may occur as there can be large variations at highest ZAMS mass (see Figure 19).

It is expected that a weaker trend will be present for that of $[^{28}Si/Fe]$. This is because some ^{28}Si is already present in the progenitor at collapse, while there is additional ^{28}Si produced by explosive oxygen burning. At the same time material in silicon rich regions can be further heated still, which can instead produce a larger amount of Fe-group material. Consequently, changes in the shock heating and mass cut can simultaneously alter both the amount of ^{28}Si and Fe-group material. Therefore the resulting trend with ZAMS mass is expected to be complex.

Finally, ^{40}Ca , which is produced primarily through incomplete silicon burning is expected to show a complex trend with ZAMS mass. This is because it is particularly sensitive to how much ejecta reaches the temperatures associated with these burning regimes (see figure 4 of Curtis et al. (2019)). A more energetic explosion can process more material into ^{40}Ca , but it can also result in more material reaching higher temperatures required for complete silicon burning and Fe-group production. Thus if Fe yield increases more rapidly than the ^{40}Ca yield, the $[^{40}Ca/Fe]$ ratio will decrease even though the absolute Ca yield has increased. We therefore expect $[^{40}Ca/Fe]$ to exhibit a relatively flat and/or complex dependence on ZAMS mass.

We show the corresponding abundance ratios in Figure 22. The $[^{16}O/Fe]$ ratio generally increases with ZAMS mass, rising from ~ 0.2 – 0.4 at $11 M_{\odot}$ to ~ 1.0 – 1.3 for the higher-mass models. Most of the exploding models between 11 and $22 M_{\odot}$ have ratios between ~ 0.2 and 0.8 , after which the increase begins to level off. The single-star and binary-stripped models exhibit broadly similar behaviour, with their ratios typically being within 0.2 at the same ZAMS mass. This similarity between the single and binary-stripped models is consistent with most of the oxygen being located in the deeper CO-rich layers of the progenitor, rather than in the outer envelope removed by binary stripping.

Despite some model-to-model scatter, we find that the $[^{28}Si/Fe]$ ratio shows a slight decrease with increasing ZAMS mass. The largest ratios occur among the lowest-mass models, reaching ~ 1.0 , before decreasing to ~ 0.1 – 0.3 by 22 – $35 M_{\odot}$. The ratio then increases slightly to ~ 0.3 – 0.4 for the highest-mass explosions. Neither progenitor series follows a smooth trend with ZAMS mass. This behaviour arises as ^{28}Si and Fe-group material are produced in neighbouring explosive-burning regimes. Relatively small changes in the shock heating cut

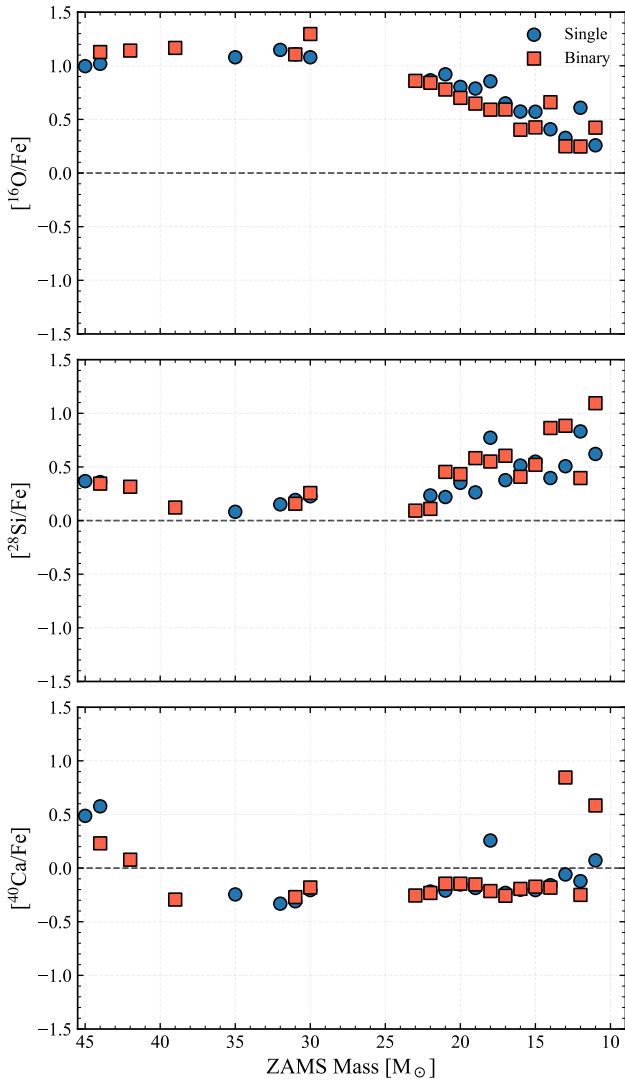


Figure 22. Top to bottom: Abundance ratios of ^{16}O , ^{28}Si and ^{40}Ca to Fe (relative to the solar ratio) for the Ms models (blue circles) and Mb models (red squares).

can therefore change how much material remains as ^{28}Si and how much is processed further into Fe-group nuclei.

As expected, $^{40}\text{Ca}/\text{Fe}$ exhibits the most scatter with ZAMS mass. Below $\sim 15 M_{\odot}$, both progenitor series contain models with positive and negative ratios, and neighbouring ZAMS masses can produce substantially different results (i.e., changing from positive to negative ratios). Most of the explosions at intermediate masses have ratios clustered around -0.2 , whereas the ratio becomes positive again for several of the highest-mass models. This behaviour is consistent with the sensitivity of ^{40}Ca and Fe-group production to that of the peak temperature and final mass cut.

6 DISCUSSION AND CONCLUSIONS

In this work, we have used the PUSH framework to investigate the explodability, remnant properties, and nucleosynthetic yields of the single-star and binary-stripped progenitors of Farmer et al. (2023). We investigated 35 single-star and 31 binary-stripped models span-

ning ZAMS masses $11\text{--}45 M_{\odot}$. The core-collapse simulations were performed using the standard PUSH calibration (Ebinger et al. 2019) and the nucleosynthesis was post-processed using the CFNET reaction network (Fröhlich et al. 2006b). Throughout this work, we compare our results with previous PUSH calculations (Ebinger et al. 2019; Curtis et al. 2019), which used single-star progenitors evolved with the KEPLER code (Woosley et al. 2002; Woosley & Heger 2007), and with the explosion calculations of Farmer et al. (2023) and Boccioli & Roberti (2026). Our main conclusions are as follows:

(i) We find that binary stripping alters the explodability landscape from $11\text{--}45 M_{\odot}$ ZAMS progenitors. However we find that there is no systematic shift towards either successful or failed explosions. Of the models investigated, 18 of the 35 single-star progenitors and 18 of the 31 binary-stripped progenitors explode successfully. The two series produce identical outcomes up to and including $22 M_{\odot}$, with all models successfully exploding. Above this mass, their outcomes begin to diverge: the first single-star failure occurs at $23 M_{\odot}$, while the first failure among the available binary-stripped models occurs at $26 M_{\odot}$. Beyond $22 M_{\odot}$, both progenitor series successfully explode at only three common ZAMS masses: 30 , 31 , and $44 M_{\odot}$. This demonstrates that binary stripping can alter the pre-collapse core structure sufficiently to change the explosion outcome at a given ZAMS mass, and while the binary-stripped models have a slightly higher explosion fraction, they are not systematically more explodable.

(ii) The predicted explodability landscape depends strongly on the adopted explosion treatment. Comparing our PUSH results with the thermal-bomb calculations of Farmer et al. (2023) and the neutrino-driven time-dependent mixing-length theory of Boccioli & Roberti (2026), we find that the approaches agree only on the outcomes of 8 of the 66 progenitor models. Moreover, $11 M_{\odot}$ is the only ZAMS mass at which all three approaches predict successful explosions for both the single-star and binary-stripped progenitors. For models that explode across the different calculations, the predicted neutron-star masses are broadly comparable, although PUSH generally produces lower neutron-star masses. We find that our predicted outcomes are also sensitive to the adopted PUSH calibration, as changing the PUSH factor can be sufficient to reverse whether a model explodes. We therefore stress that our explodability landscape depends on the assumption that compactness is the appropriate quantity with which to calibrate the additional PUSH heating.

(iii) We find that the successfully exploding single-star and binary-stripped models occupy broadly similar regions of explosion parameter space. Their explosion energies are $\sim 0.65\text{--}1.3 \text{ B}$, their neutron-star masses are $\sim 1.5\text{--}2.0 M_{\odot}$, and their explosion times are $\sim 0.35\text{--}0.65 \text{ s}$. These values are consistent with previous PUSH calculations (Ebinger et al. 2019) using the WHW02 and WH07 progenitors (Woosley et al. 2002; Woosley & Heger 2007). We stress that the differences in these quantities introduced by binary stripping are comparable to the scatter between the WHW02 and WH07 progenitor sets, which were evolved using nuclear reaction networks of different sizes. As in Ebinger et al. (2019), the explosion properties show clearer trends with compactness rather than with ZAMS mass. The most compact progenitors generally produce the most massive neutron stars, later and weaker explosions, while models with intermediate compactness produce the most energetic explosions, and the least compact models produce the lowest neutron-star masses and weaker explosions.

(iv) After nucleosynthesis post-processing for the successful explosions, we find that the largest systematic differences between the single-star and binary-stripped yields occur for the lightest elements. The single-star models eject more light elements simply because

they retain their extended envelopes. However, when the yields are normalised to either the corresponding pre-collapse abundances or the amount of Fe produced by each explosion differences emerge between the two series. Between ~ 11 and $14 M_{\odot}$, binary stripping produces substantial differences among the elements from S to Sc-material which is produced through explosive O and Si burning. While we find that the Fe-group yields also exhibit scatter in the yields, this reflects differences in the Si-rich structure close to the mass cut. At higher ZAMS masses, the elemental yields of the two progenitor series are generally more comparable.

(v) Using a Salpeter IMF weighting, we find that the single-star and binary-stripped populations produce broadly similar elemental yield patterns, with most elements differing by only ~ 20 per cent. Relative to the single-star population, the binary-stripped population shows moderate enhancements in Cl, Ar, and Sc, while N, F, Cu, and several Fe-group elements are suppressed. The clearest difference occurs for K which is enhanced by a factor of ~ 3.4 in the binary-stripped population. This enhancement is driven by the increased production of ^{39}K in the lowest-mass progenitors and is amplified by the greater weight assigned to lower-mass models by the Salpeter IMF weighting. We find that this enhancement is consistent with that of Farmer et al. (2023).

(vi) Overall, the predicted nucleosynthetic yields are broadly consistent with observational constraints. Our ^{56}Ni yields span much of the observed CCSN range for progenitor masses below $\sim 23 M_{\odot}$, while the iron-group abundance patterns generally reproduce those observed in the metal-poor star HD 84937, aside from substantial model-to-model variation in Sc and Zn and a systematic overproduction of Cu. The α -to-iron ratios also follow the expected behaviour with ZAMS mass, with $[^{16}\text{O}/\text{Fe}]$ increasing most clearly, $[^{28}\text{Si}/\text{Fe}]$ showing a weaker trend, and $[^{40}\text{Ca}/\text{Fe}]$ remaining particularly sensitive to the explosion conditions and mass cut. Despite this variation, none of these abundance ratios shows a systematic separation between the single-star and binary-stripped progenitors.

ACKNOWLEDGEMENTS

We acknowledge Numpy (Harris et al. 2020), SciPy (Virtanen et al. 2020) and Matplotlib (Harris et al. 2020).

DATA AVAILABILITY

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APPENDIX A: COMPACTNESS AS A FUNCTION OF ZAMS MASS

Figure A1 shows the compactness evaluated at $2.0 M_{\odot}$ at bounce, $\xi_{2.0}$, which is used to determine the additional PUSH heating through k_{push} , as a function of ZAMS mass (see Ebinger et al. 2019, for more details).

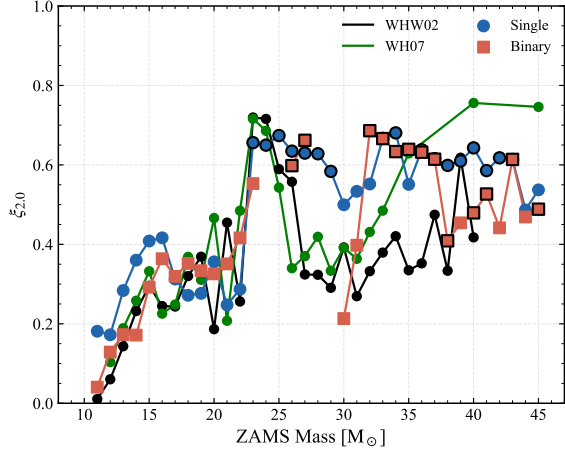


Figure A1. $\xi_{2,0}$ compactness at core bounce as a function of ZAMS mass for the single-star and binary-stripped progenitors. Solar-metallicity progenitors from [Woosley et al. \(2002\)](#); [Woosley & Heger \(2007\)](#) are shown for comparison.

APPENDIX B: ISOTOPIC YIELDS OF THE MS18.00 AND MB18.00 MODELS

Table B1 presents the isotopic yields of the Ms18.00 and Mb18.00 models. These models are used throughout Section 4 to illustrate the differences in nucleosynthesis between the single-star and binary-stripped progenitors. The table includes isotopic yields through the end of the Ni isotopes for the Ms18.00 and Mb18.00 models. The remaining isotopic yields for these two models, together with the complete yield tables for all other successfully exploding models, are provided electronically.

Table B1: Pre-explosion, net change, and final isotopic yields, in units of $M_{\odot}$, for the representative Ms18.00 and Mb18.00 models. The net change is defined as the final isotopic mass minus the corresponding pre-supernova mass after all tracers have cooled below 1.75 GK. Negative and positive values therefore indicate a decrease or increase, respectively, in the total mass of a given isotope over the course of collapse and explosion. Note that these net changes can also reflect the reprocessing of material during collapse and explosion, such that substantial destruction and subsequent re-production of an isotope may result in only a small net change in its final abundance.

Isotope	Ms18			Mb18		
	Pre-explosion	Explosive	Final	Pre-explosion	Explosive	Final
¹ H	1.76E+00	-4.71E-04	1.76E+00	7.28E-04	-5.52E-04	1.76E-04
² H	9.64E-09	-9.64E-09	4.58E-17	1.57E-08	-1.57E-08	7.09E-20
³ He	5.16E-05	-2.98E-11	5.16E-05	5.22E-11	-5.08E-11	1.37E-12
⁴ He	2.67E+00	-2.11E-03	2.67E+00	1.28E+00	-3.66E-03	1.28E+00
⁶ Li		1.36E-19	1.36E-19		5.52E-20	5.52E-20
⁷ Li	2.41E-12	1.19E-11	1.43E-11	3.72E-12	3.32E-11	3.69E-11
⁹ Be	4.03E-11	-1.87E-12	3.84E-11	3.20E-12	-3.20E-12	1.66E-19
¹⁰ B	1.99E-10	-1.42E-13	1.99E-10	3.44E-13	-3.43E-13	1.65E-16
¹¹ B	8.92E-10	6.01E-11	9.52E-10	1.59E-12	2.13E-11	2.29E-11
¹² C	3.06E-01	7.05E-05	3.07E-01	2.46E-01	-6.36E-04	2.45E-01
¹³ C	2.18E-04	3.24E-06	2.21E-04	2.85E-05	8.12E-07	2.94E-05
¹⁴ N	1.02E-02	2.72E-05	1.02E-02	4.69E-03	6.67E-06	4.69E-03
¹⁵ N	7.69E-06	1.39E-05	2.16E-05	4.23E-06	3.22E-06	7.45E-06
¹⁶ O	2.19E+00	-3.60E-02	2.16E+00	1.54E+00	-4.57E-02	1.50E+00
¹⁷ O	1.68E-05	2.44E-06	1.93E-05	9.15E-06	7.22E-07	9.87E-06
¹⁸ O	1.17E-03	2.01E-05	1.19E-03	6.79E-04	1.76E-05	6.97E-04
¹⁹ F	3.50E-06	2.45E-07	3.75E-06	3.77E-06	5.46E-08	3.82E-06
²⁰ Ne	9.56E-01	-3.96E-03	9.52E-01	3.79E-01	-2.80E-02	3.51E-01
²¹ Ne	2.07E-03	8.14E-04	2.89E-03	1.68E-04	6.76E-04	8.44E-04
²² Ne	2.71E-02	1.24E-03	2.84E-02	1.52E-02	1.08E-03	1.63E-02
²³ Na	2.82E-02	-1.61E-03	2.65E-02	1.51E-03	1.42E-04	1.65E-03
²⁴ Mg	9.37E-02	-3.05E-04	9.34E-02	2.50E-01	-7.03E-03	2.43E-01
²⁵ Mg	2.40E-02	-1.87E-03	2.21E-02	1.54E-02	-1.35E-03	1.41E-02
²⁶ Mg	2.13E-02	-1.60E-03	1.97E-02	1.15E-02	-4.74E-04	1.10E-02
²⁷ Al	1.69E-02	-2.18E-03	1.48E-02	2.21E-02	-4.72E-03	1.74E-02
²⁸ Si	2.94E-01	-1.04E-01	1.91E-01	3.24E-01	-1.78E-01	1.46E-01
²⁹ Si	2.37E-03	1.46E-03	3.83E-03	1.43E-02	-2.17E-03	1.21E-02
³⁰ Si	2.21E-03	3.70E-04	2.58E-03	1.15E-02	-1.09E-03	1.04E-02
³¹ P	2.89E-03	-1.07E-03	1.83E-03	3.75E-03	-1.29E-03	2.45E-03
³² S	1.29E-01	-4.44E-02	8.49E-02	8.64E-02	-5.90E-02	2.75E-02
³³ S	1.57E-03	2.03E-04	1.78E-03	9.36E-04	-5.66E-04	3.70E-04
³⁴ S	8.81E-03	-4.26E-03	4.55E-03	8.38E-03	-6.19E-03	2.19E-03
³⁶ S		9.89E-05	9.89E-05		5.96E-06	5.96E-06
³⁵ Cl	2.61E-03	-2.06E-03	5.50E-04	1.29E-03	-1.10E-03	1.91E-04
³⁷ Cl	1.62E-04	1.60E-04	3.22E-04	9.49E-05	-4.66E-05	4.83E-05
³⁶ Ar	1.92E-02	-7.79E-03	1.14E-02	1.11E-02	-7.82E-03	3.28E-03
³⁸ Ar	1.01E-02	-8.21E-03	1.92E-03	6.68E-03	-6.02E-03	6.68E-04
⁴⁰ Ar	2.36E-05	6.90E-06	3.05E-05	2.61E-05	-2.26E-05	3.50E-06
³⁹ K	2.56E-03	-2.35E-03	2.07E-04	8.89E-04	-8.34E-04	5.50E-05
⁴¹ K	1.14E-05	8.18E-05	9.33E-05	1.20E-05	-7.05E-06	4.99E-06
⁴⁰ Ca	9.75E-03	-3.84E-03	5.91E-03	6.51E-03	-3.96E-03	2.55E-03
⁴² Ca	4.48E-04	-3.63E-04	8.49E-05	2.28E-04	-2.10E-04	1.76E-05
⁴³ Ca	3.04E-05	-9.03E-06	2.14E-05	3.59E-05	-3.32E-05	2.77E-06
⁴⁴ Ca	1.40E-04	-6.38E-05	7.60E-05	1.70E-04	-1.25E-04	4.44E-05
⁴⁶ Ca	1.01E-03	-9.96E-04	1.55E-05	8.82E-04	-8.82E-04	6.48E-07
⁴⁵ Sc	4.90E-05	-3.17E-05	1.72E-05	6.87E-05	-6.67E-05	2.02E-06

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Table B1 – continued from previous page

Isotope	Ms18			Mb18		
	Pre-explosion	Explosive	Final	Pre-explosion	Explosive	Final
⁴⁶ Ti	4.22E-04	-3.95E-04	2.62E-05	4.80E-04	-4.72E-04	7.53E-06
⁴⁷ Ti	3.37E-04	-3.25E-04	1.23E-05	4.58E-04	-4.55E-04	3.86E-06
⁴⁸ Ti	3.12E-03	-3.02E-03	1.00E-04	3.97E-03	-3.86E-03	1.13E-04
⁴⁹ Ti	5.69E-03	-5.69E-03	8.05E-06	6.39E-03	-6.39E-03	5.81E-06
⁵⁰ Ti	5.89E-02	-5.89E-02	1.07E-05	5.14E-02	-5.14E-02	5.44E-06
⁵¹ V	3.30E-02	-3.30E-02	1.25E-05	3.71E-02	-3.71E-02	1.24E-05
⁵⁰ Cr	9.41E-03	-9.36E-03	5.43E-05	1.01E-02	-1.00E-02	5.14E-05
⁵² Cr	6.61E-02	-6.55E-02	6.54E-04	7.83E-02	-7.75E-02	7.46E-04
⁵³ Cr	4.49E-02	-4.48E-02	7.12E-05	5.10E-02	-5.09E-02	7.78E-05
⁵⁴ Cr	1.10E-01	-1.10E-01	2.16E-05	1.03E-01	-1.03E-01	1.32E-05
⁵⁵ Mn	6.68E-02	-6.65E-02	3.74E-04	7.60E-02	-7.56E-02	3.84E-04
⁵⁴ Fe	1.45E-01	-1.42E-01	3.38E-03	1.38E-01	-1.34E-01	3.50E-03
⁵⁶ Fe	1.56E-01	-9.43E-02	6.16E-02	1.73E-01	-9.35E-02	7.97E-02
⁵⁷ Fe	6.13E-02	-5.90E-02	2.23E-03	6.95E-02	-6.65E-02	3.02E-03
⁵⁸ Fe	9.78E-02	-9.69E-02	9.00E-04	9.64E-02	-9.57E-02	7.44E-04
⁵⁹ Co	2.03E-02	-1.97E-02	5.97E-04	2.31E-02	-2.27E-02	4.29E-04
⁵⁸ Ni	3.37E-02	-3.09E-02	2.78E-03	3.53E-02	-3.08E-02	4.53E-03
⁶⁰ Ni	1.73E-02	-1.57E-02	1.67E-03	1.93E-02	-1.72E-02	2.06E-03
⁶¹ Ni	1.06E-02	-1.04E-02	2.00E-04	1.21E-02	-1.19E-02	1.54E-04
⁶² Ni	2.13E-02	-2.02E-02	1.06E-03	2.17E-02	-2.07E-02	1.02E-03
⁶⁴ Ni	2.16E-02	-2.13E-02	3.03E-04	1.43E-02	-1.42E-02	1.70E-04
⁶³ Cu	8.58E-04	-7.77E-04	8.11E-05	9.55E-04	-9.15E-04	3.99E-05
⁶⁵ Cu		4.46E-06	4.46E-06		6.56E-06	6.56E-06
⁶⁴ Zn	4.77E-04	-9.16E-05	3.85E-04	2.19E-04	-1.33E-05	2.05E-04

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